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1. A blue-flowered African violet of unknown ancestry self-pollinated and produced 50 seeds. These seeds germinate and grow into flowering plants. Of these plants, 36 produce blue flowers and 14 produce pink flowers. What is the best explanation for the pink-flowered offspring?

(A) Blue flowers are incompletely dominant to pink flowers.

(B) Pink flower color is a trait recessive to blue flower color. ✓

(C) Pink flower color is the result of somatic mutations in the flower color gene.

(D) A previous generation of the blue-flowered parent must have included 50 percent pinkflowered plants.

2. Several members of a family are found to involuntarily sneeze when exposed to bright lights, such as sunlight. Following analysis of the condition in the family, a doctor predicts that the symptoms have an underlying genetic basis. A pedigree of the family is shown in Figure 1.

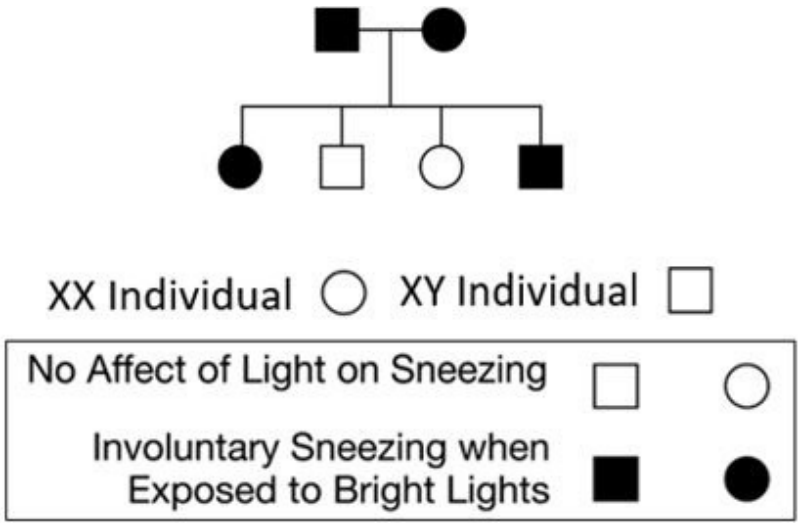


Figure 1. Pedigree showing family members with and without symptoms

For this condition, which of the following modes of inheritance is most consistent with the observations?

(A) Autosomal dominant ✓

(B) Autosomal recessive

(C) X-linked dominant

(D) X-linked recessive

Answer A

Correct. Based on the pedigree, the most likely explanation is autosomal dominant inheritance. Both

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parents are likely heterozygous for the condition, making it possible to have two children that are not affected. The unaffected children have a homozygous recessive genotype.

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- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

The three-spined stickleback (*Gasterosteus aculeatus*) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mainly of individuals with armor-like plates covering most of their body surface (completely plated). Approximately 10,000 years ago, some marine sticklebacks colonized freshwater environments. After many generations in the freshwater environments, the freshwater stickleback populations lacked the armor plating (low plated) typical of marine stickleback populations.

Over the period between 1957 and 2005, one freshwater population, in Lake Washington, a lake in a coastal region of the northwestern United States, changed from having a majority of individuals of the low-plated phenotype to having more individuals of the completely-plated phenotype than of the low-plated phenotype. Figure 1 shows the distribution of plated phenotypes in Lake Washington sticklebacks at four time points between 1957 and 2005.

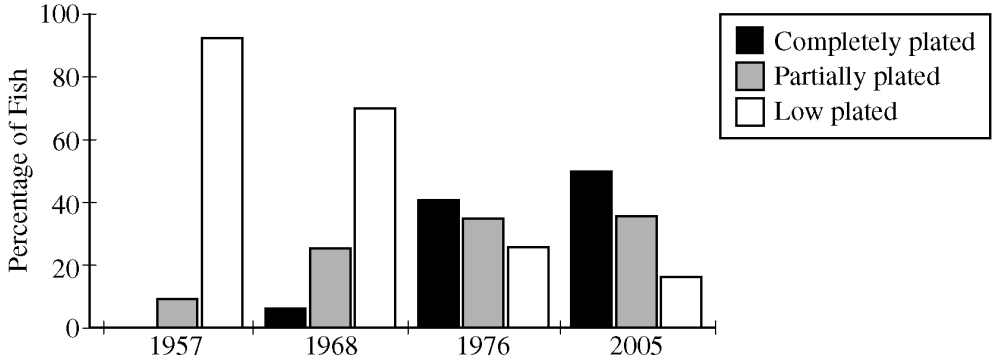


Figure 1. Armor plating phenotypes in Lake Washington stickleback population

A single gene, *ectodysplasin* (*EDA*), is thought to be responsible for the variation in the number of armor plates in sticklebacks. Figure 2 shows a phylogenetic tree constructed by comparing DNA sequences of the *EDA* gene from a number of stickleback populations with low-plated or completely plated phenotypes. Figure 3 shows a phylogenetic tree constructed by comparing the sequences of 25 genes that were randomly selected from the same populations as shown in Figure 2. In both figures, shaded populations display the completely plated phenotype.

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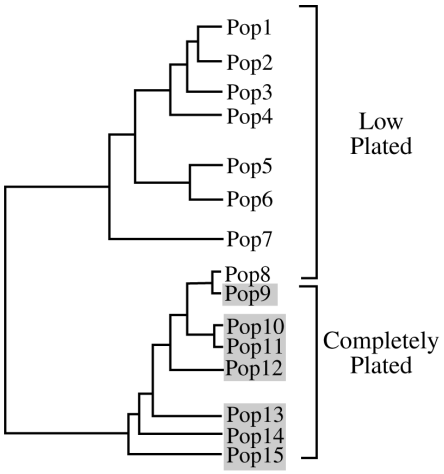


Figure 2. Phylogeny based on *EDA* gene only

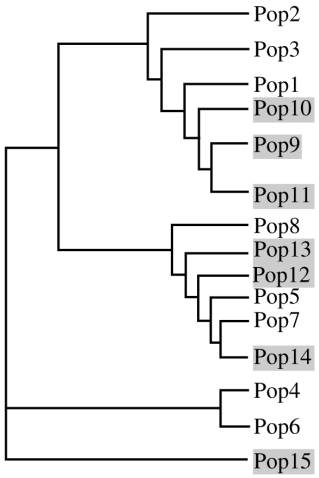


Figure 3. Phylogeny based on 25 random genes

3. A completely-plated stickleback from a marine population was mated to a low-plated stickleback from a freshwater population. The resulting F_1 hybrids all displayed a completely plated phenotype. When the F_1 hybrids were allowed to interbreed, the resulting F_2 generation included completely plated offspring and low-plated offspring in an approximate 3:1 ratio. Which of the following conclusions is best supported by the results of the breeding experiments?

(A) Phenotypic variation in the F_2 generation suggests that armor morphology is controlled by many alleles of a single gene.

(B) The completely-plated phenotype is controlled by a dominant allele of a single gene. ✓

(C) Armor loss is an acquired characteristic that is affected by one or more environmental factors.

(D) Patterns of armor plating in stickleback populations are regulated by sex-specific signals.

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4. A couple has 5 children, all sons. If the woman gives birth to a sixth child, what is the probability that the sixth child will be a son?

(A) 5/6

(B) 1/2 ✓

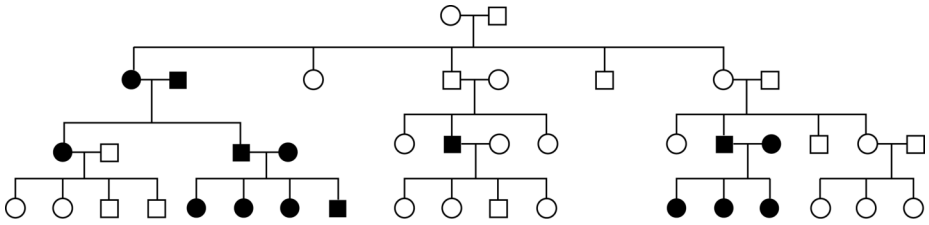
(C) 1/5

(D) 1/6

(E) 1/64

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5.



In the pedigree above, circles represent females, squares represent males, and shaded figures represent individuals expressing a specific trait. The expression of this trait is most likely due to which of the following?

(A) Sex-linked dominant inheritance

(B) Sex-linked recessive inheritance

(C) Autosomal dominant inheritance

(D) Autosomal recessive inheritance ✓

(E) A codominant relationship of a single pair of alleles

6. An African violet grower observes that genetically identical African violet plants growing near the walls of the greenhouse have white flowers, that plants growing farther away from the walls have pale blue flowers, and that plants growing nearest the center of the greenhouse have dark blue flowers.

Which of the following best explains the differences in flower color of the African violets in the greenhouse?

(A) Warmer temperatures result in genotypic alterations, which result in flower color differences.

(B) The plants along the walls of the greenhouse are homozygous recessive and therefore have white flowers.

(C) An enzyme responsible for flower color does not fold correctly in cooler temperatures, and the greenhouse is warmest in the center. ✓

(D) More light is available along the walls of the greenhouse, so the flowers need less pigment to absorb sunlight for photosynthesis.

Answer C

Correct. If an enzyme responsible for pigment production does not fold correctly in cooler temperatures, then the enzyme's active site will not be in the correct shape and it will not be able to catalyze the required reaction effectively. If temperatures are cooler along the walls of the green house, then the flowers of plants along the walls will have less pigment.

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7. A genetic counselor is consulted by a young man who is worried about developing Huntington's disease, an inherited disorder caused by a dominant allele of a single gene. The young man explains that his cousin was recently diagnosed with Huntington's disease, and the news has caused him to consider his own risk of developing the disorder. Which of the following questions will best help the genetic counselor to evaluate the risk of the young man developing Huntington's disease and transmitting it to his children?

(A) Were you and your cousin born in the same geographical area?

(B) Were your parents or grandparents ever diagnosed with Huntington's disease? ✓

(C) Were you in physical contact with a person diagnosed with Huntington's disease?

(D) Were you ever exposed to substances that are suspected of being mutagens?

Directions: Each group of questions below concerns an experimental or a laboratory situation. In each case, first study the description of the situation. Then choose the one best answer to each question following it.

In dogs, one pair of alleles determines coat color (dark and albino). Another pair of alleles determines hair length (short and long). Thus, each gamete will contain one of the coat-color alleles, *C* or *c* and one of the hair-length alleles, *B* or *b*. In repeated crosses of a specific dark, short-haired dog with an albino, long-haired dog, all the offspring were dark with short hair, as shown in cross I. However, in subsequent crosses of another dark, short-haired dog with a dark, long-haired dog, the ratios shown in cross II below were obtained.

Cross	*Parents*	*Offspring*
I.	Dark, short hair x Albino, long hair	All dark, short hair
II.	Dark, short hair x Dark, long hair	3 dark, short hair 3 dark, long hair 1 albino, short hair 1 albino, long hair

8. In cross II, the genotype of the dark, short-haired parent is

(A) *CcBb* ✓

(B) *ccbb*

(C) *CCBB*

(D) *CCbb*

(E) *ccBB*

9. Which of the following is probably the genotype of the dark, short-haired parent in cross I?

(A) *CcBb*

(B) *ccbb*

(C) *CCBB* ✓

(D) *CCbb*

(E) *ccBB*

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10. Which of the following correctly describes the relationship of the dark-coat-color allele to the albino condition?

(A) It is dominant. ✓

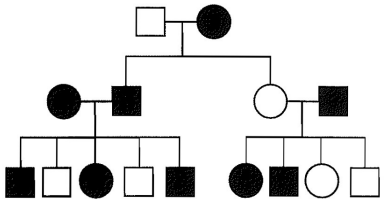
(B) It is recessive.

(C) It is codominant.

(D) It is a polygenic inheritance pattern.

(E) The alleles are linked.

11.



In the pedigree above, squares represent males and circles represent females. Individuals who express a particular trait are represented by shaded figures. Which of the following patterns of inheritance best explains the transmission of the trait?

(A) Sex-linked dominant

(B) Sex-linked recessive

(C) Autosomal recessive

(D) Autosomal dominant ✓

(E) Incompletely dominant

12. A series of crosses is performed with fruit flies (*Drosophila melanogaster*) to examine inheritance of the genes *vestigial* (*vg*) and *cinnabar* (*cn*). The recessive *vg* allele causes small, malformed wings called vestigial wings. The recessive *cn* allele causes bright-red eyes called cinnabar eyes.

In the first cross, a female with wild-type wings and eyes is mated with a male with vestigial wings and cinnabar eyes. All the F_1 individuals have wild-type wings and eyes. In the second cross, female F_1 flies are mated with males with vestigial wings and cinnabar eyes. The phenotypes of 500 F_2 individuals are shown in the table

Phenotype	Number of Individuals
Wild-type wings, wild-type eyes	226
Wild-type wings, cinnabar eyes	25
Vestigial wings, wild-type eyes	26
Vestigial wings, cinnabar eyes	223

Which of the following is the most likely explanation of the results?

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(A) The two genes are located on two different chromosomes.

(B) The two genes are sex-linked.

(C) The two genes are located on mitochondrial DNA.

(D) The two genes are linked on an autosome. ✓

13. Assume that genes *A* and *B* are not linked. If the probability of allele *A* in a gamete is 1/2 and the probability of allele *B* in a gamete is 1/2, then the probability that both *A* and *B* are in the same gamete is

(A) 1/2 x 1/2 ✓

(B) 1/2 + 1/2

(C) (1/2) / (1/2)

(D) $\sqrt{1/2}$

(E) |1/2 — 1/2|

14. A student in a biology class crossed a male *Drosophila melanogaster* having a gray body and long wings with a female *D. melanogaster* having a black body and apterous wings. The following distribution of traits was observed in the offspring.

Phenotype	Number of Offspring
Gray body, long wings	42
Black body, apterous wings	41
Gray body, apterous wings	9
Black body, long wings	8

Which of the following is supported by the data?

(A) The alleles for gray body and long wings are dominant.

(B) The alleles for gray body and long wings are recessive.

(C) Genes for the two traits are located on two different chromosomes, and independent assortment occurred.

(D) Genes for the two traits are located close together on the same chromosome, and crossing over occurred between the two gene loci. ✓

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Table I shows the results of breeding experiments to examine the inheritance of flower color (purple versus white) and pod shape (inflated versus constricted). For the crosses recorded in Table I, true-breeding parents were crossed to produce F_1 offspring, which were then testcrossed to homozygous recessive individuals. Table II shows the results of computer-simulated crosses to model the inheritance of leaf shape (broad versus narrow) and flower color (purple versus white).

TABLE I: RESULTS FROM CROSSES WITH PEA PLANTS

Parental Cross	Phenotypes of F_1 Offspring	Phenotypes of Testcross Offspring (numbers of individuals)			
Purple × White	Purple	Purple (461)		White (468)	
Inflated × Constricted	Inflated	Inflated (593)		Constricted (588)	
Purple, Inflated × White, Constricted	Purple, Inflated	Purple, Inflated (315)	Purple, Constricted (312)	White, Inflated (320)	White, Constricted (317)

TABLE II: RESULTS OF COMPUTER-SIMULATED CROSSES

Parental Cross	Phenotypes of F_1 Offspring	Phenotypes of Testcross Offspring (numbers of individuals)			
Broad × Narrow	Broad	Broad (4870)		Narrow (4862)	
Purple × White	Purple	Purple (4253)		White (4259)	
Broad, White × Narrow, Purple	Broad, Purple	Broad, White (672)	Broad, Purple (75)	Narrow, White (61)	Narrow, Purple (664)

15. Based on the data in Table I, which of the following best explains why there are no individuals with constricted pods in the F_1 generation?

(A) Inflated pod shape is dominant to constricted pod shape. ✓

(B) The inflated-pod offspring in the F_1 generation are homozygous.

(C) Constricted pod shape typically arises from a new mutation in the F_1 generation.

(D) The constricted-pod offspring are carriers for the inflated pod shape allele.

16. In Table I, the ratio of phenotypes in the offspring from the testcross with F_1 plants that had purple flowers and inflated pods suggests that the genes for flower color and pod shape are located

(A) close together on the same autosome

(B) on the X chromosome

(C) on different chromosomes ✓

(D) on a mitochondrial chromosome

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17. Which of the following provides the best justification for an assumption that might have been used in the computer simulation (Table II)?

(A) The broad allele is recessive to the narrow allele because broad leaves appear in every generation.

(B) The purple allele is dominant to the white allele because all the offspring from the cross of purple-flowered and white-flowered plants had purple flowers. ✓

(C) The narrow allele is codominant with the purple allele because the purple-flower trait and the narrow-leaf trait segregate together.

(D) The white allele is dominant to both the broad and narrow alleles because plants with either type of leaf shape can have white flowers.

18. In Table II, the F_1 offspring of the cross between broad-leaved, white-flowered plants with narrow-leaved, purple-flowered plants have a phenotype that differs from that of either parent. However, many testcross offspring have the same phenotype as one of the two plants in the parental cross, but relatively few testcross offspring have the same phenotype as the F_1 offspring. Which of the following best explains the observation?

(A) Recombination between the leaf-shape and flower-color genes resulted in chromosomes carrying a dominant allele of both genes. ✓

(B) Recombination between the broad and narrow alleles of the leaf-shape gene resulted in chromosomes carrying three different alleles at the same genetic locus.

(C) Independent assortment of homologous chromosomes resulted in the combinations of alleles present in the parental generation.

(D) The computer model cannot capture the possible assortments of gametes when multiple genes are considered.

19.

	Cross 1		Cross 2		Cross 3	
Parents	Blue Male × Blue Female		Blue Male × White Female		White Male × Blue Female	
Offspring	Males	Females	Males	Females	Males	Females
	60 blue	124 blue	0 blue	115 blue	48 blue	42 blue
	52 white	0 white	108 white	0 white	53 white	49 white

The data above represent the results of three different crosses involving the inheritance of a gene that determines whether a certain organism is blue or white. Which of the following best explains the mechanism of inheritance of the gene?

(A) The allele for white is an autosomal dominant allele because a $1:1$ phenotype ratio of blue to white among both sexes is observed in cross 3.

(B) The allele for blue is an autosomal dominant allele because an approximate $3:1$ phenotype ratio of blue to white is observed in cross 1.

(C) The allele for white is an X-linked dominant allele because no white females are produced in cross 1.

(D) The allele for blue is an X-linked dominant allele because there are no blue male offspring in cross 2. ✓

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20. Butterflies of the genus Colias live in the Rocky Mountains, where they experience a wide range of temperatures. Different variants of a particular glycolytic enzyme in the flight muscles are optimally active at different temperatures. Within the same population, some individual butterflies fly most effectively at 29°C , while others fly most effectively at 40°C. Still others can be equally active at both temperatures. Which of the following claims is most consistent with the observed butterfly behavior?

(A) Butterflies that express two variants of the enzyme are active over a greater range of temperature. ✓

(B) Butterflies that are active over a wide range of temperatures produce greater amounts of the enzyme.

(C) Temperature has little effect on the activity of butterflies.

(D) Butterflies that are active at warmer temperatures produce more offspring.

21. A scientist studying phenotypic variation in a species of butterfly observed that genetically identical caterpillars grown in similar cages but exposed to different colored lights developed into butterflies with differences in wing color and body size, as shown in Table 1.

Table 1. Effect of Exposing Identical Caterpillars to Specific Colors of Light

Phenotype of Adult Butterfly	Caterpillars Exposed to Red Light	Caterpillars Exposed to Blue Light
Wing color	Darker	Lighter
Body size	Smaller	Larger

Which of the following best explains the cause of the phenotypic variation observed in the butterflies?

(A) Different mutations occurred in the caterpillars that were exposed to different colors of light.

(B) The energy used to grow a larger body results in butterflies with lighter colored wings.

(C) Individual caterpillars evolved adaptations to survive in each of the light conditions they were exposed to.

(D) There was differential gene expression of wing color and body size in response to the colors of light the caterpillars were exposed to. ✓

Answer D

Correct. Phenotype is a product of both genes and environment. Since the butterflies are genetically identical, their phenotype differences must be a product of the environment interacting with the genotype.

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22. A gene that influences coat color in domestic cats is located on the X chromosome. A female cat that is heterozygous for the gene ($\mathrm{X}^{B}\mathrm{X}^{O}$) has a calico-colored coat. In a genetics experiment, researchers mate a calico-colored female cat ($\mathrm{X}^{B}\mathrm{X}^{O}$) with an orange-colored male cat ($\mathrm{X}^{O}\mathrm{Y}$) to produce an F_1 generation. The researchers record observations for the cats in the F_1 generation and plan to use the data to perform a chi-square goodness-of-fit test for a model of X-linked inheritance. The data for the chi-square goodness-of-fit test are presented in Table 1.

Table 1. Data for the chi-square goodness-of-fit test

Phenotype	Genotype	Observed	Expected
Calico-colored female	$\mathrm{X}^{B}\mathrm{X}^{O}$	15	10
Orange-colored female	$\mathrm{X}^{O}\mathrm{X}^{O}$	6	10
Black-colored male	$\mathrm{X}^{B}\mathrm{Y}$	11	10
Orange-colored male	$\mathrm{X}^{O}\mathrm{Y}$	8	10

The researchers calculate a chi-square value of 4.6 and choose a significance level of $p = 0.05$. Which of the following statements best completes the chi-square goodness-of-fit test?

(A) The null hypothesis can be rejected because the chi-square value is greater than the critical value.

(B) The null hypothesis can be rejected because the chi-square value is less than the critical value.

(C) The null hypothesis cannot be rejected because the chi-square value is greater than the critical value.

(D) The null hypothesis cannot be rejected because the chi-square value is less than the critical value. ✓

Answer D

Correct. There are four potential phenotypic categories, so there are three degrees of freedom. At the $p = 0.05$ significance level, the critical value is 7.82. Because the calculated chi-square value is less than the critical value, the null hypothesis cannot be rejected.

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23.

TABLE 1. NUMBER OF CHICKENS OF EACH PHENOTYPE

Phenotype	Number
Black, solid	47
Black, spotted	12
Gray, solid	16
Gray, spotted	4
Total	79

In a certain type of chicken, the allele for gray feathers is recessive to the allele for black feathers, and the allele for a spotted pattern is recessive to the allele for a solid pattern. Chickens that are heterozygous for both traits were crossed and the phenotypes for the offspring were recorded (Table 1).

To determine whether the data are consistent with the expectations for independent assortment, students performed a chi-square analysis.

Which of the following is the closest to the calculated chi-square value for the experiment?

(A) -0.26

(B) 0.20

(C) 0.95 ✓

(D) 1.25

Answer C

Correct. Chi-square is calculated by first determining the expected phenotype values based on the phenotype ratio of $9:3:3:1$ for a dihybrid cross with unlinked genes. Then the observed values provided in Table 1 and the expected values are plugged into the chi-square formula. The expected number of black, solid chickens is $79 \times (9/16) = 44.44$. The expected number of black, solid chickens and of gray, solid chickens is $79 \times (3/16) = 14.81$ each. The expected number of gray, spotted chickens is $79 \times (1/16) = 4.94$. Using these values in the chi-square formula yields 0.95.

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24. In fruit flies of the genus *Drosophila*, the allele for vestigial wings is recessive to the allele for round wings, and the allele for brown eye color is recessive to the allele for red eye color. A scientist crossed flies that are heterozygous for both traits and determined the number of offspring with each combination of phenotypes. The scientist performed a chi-square analysis to determine if the data are consistent with the expectations for independent assortment. The chi-square calculated value for the experiment was 6.03.

Based on the chi-square calculated value, which of the following statements is most accurate?

(A) The chi-square calculated value is greater than the chi-square critical value; therefore the null hypothesis should be rejected.

(B) The chi-square calculated value is greater than the chi-square critical value; therefore the null hypothesis should not be rejected.

(C) The chi-square calculated value is less than the chi-square critical value; therefore the null hypothesis should be rejected.

(D) The chi-square calculated value is less than the chi-square critical value; therefore the null hypothesis should not be rejected. ✓

Answer D

Correct. Degrees of freedom (df) equals $n - 1$, where n is the number of different categories, or in this example, phenotypes, into which the data are divided. Degrees of freedom is important because a greater number of categories means a greater likelihood of deviation from the expected value. The chi-square critical value for three degrees of freedom ($\mathrm{df} = 3$) and $\mathrm{p} > 0.05$ is 7.81. Because the calculated value 6.03 is less than 7.81, the observed frequencies do not significantly deviate from the expectations for independent assortment, and the null hypothesis is not rejected.

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25. Rubber rabbitbrush plants display heritable variation in plant height and insect-induced gall formation. In a field study, researchers investigated the relationship between plant height and gall number for the rubber rabbitbrush plants in a natural population. The researchers used the data they collected to perform a chi-square test of independence. The null hypothesis for the chi-square test was that plant height and gall number are independent. The data for the chi-square test are presented in Table 1.

Table 1. Data for the chi-square test of independence

Height Category (cm)	Number of Plants with Ten or Fewer Galls	Number of Plants with More than Ten Galls
0–30	34	6
31–60	22	16
61–90	14	21
91–120	10	27

The researchers calculated a chi-square value of 29.25. If there are three degrees of freedom and the significance level is $p = 0.05$, which of the following statements best completes the chi-square test?

(A) The critical value is 0.05, and the null hypothesis cannot be rejected because the calculated chi-square value is greater than the critical value.

(B) The critical value is 0.05, and the null hypothesis can be rejected because the calculated chi-square value is greater than the critical value.

(C) The critical value is 7.82, and the null hypothesis cannot be rejected because the calculated chi-square value is greater than the critical value.

(D) The critical value is 7.82, and the null hypothesis can be rejected because the calculated chi-square value is greater than the critical value. ✓

Answer D

Correct. The critical value is 7.82. Because the calculated chi-square value is greater than the critical value, the null hypothesis can be rejected.

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26. Researchers performed a dihybrid cross with coffee bean plants to investigate whether the inheritance of two traits (height and stem circumference) follows Mendel's law of independent assortment. The data for the F_2 generation are presented in Table 1.

Table 1. Data for the F_2 generation

Phenotype	Observed	Expected
Dwarf with small stem circumference	191	172
Dwarf with large stem circumference	26	57
Tall with small stem circumference	39	57
Tall with large stem circumference	49	19
TOTAL	305	305

Which of the following is closest to the calculated chi-square (χ^2) value for the data presented in Table 1?

(A) 8.35

(B) 72.01 ✓

(C) 98.00

(D) 2,546.00

Answer B

Correct. 72.01 is the correct calculated answer using the formula $\chi^2 = \Sigma \frac{(o-e)^2}{e}$.

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27. In corn plants, purple kernel color is dominant to yellow kernel color, and smooth kernels are dominant to wrinkled kernels. Researchers carried out a genetics experiment to investigate whether the genes controlling kernel color and kernel texture segregate independently. In their experiment, the researchers crossed two corn plants that were each heterozygous for both kernel color and kernel texture. The results of the experiment are presented in Table 1.

Table 1. Results of a genetic cross between two heterozygous corn plants

Offspring Phenotype	Observed	Expected
Purple, smooth kernels	918	900
Purple, wrinkled kernels	281	300
Yellow, smooth kernels	314	300
Yellow, wrinkled kernels	87	100

Using a significance level of $p = 0.05$, which of the following statements best completes a chi-square goodness-of-fit test for a model of independent assortment?

(A) The calculated chi-square value is 0.66, and the critical value is 0.05. The null hypothesis can be rejected.

(B) The calculated chi-square value is 0.66, and the critical value is 3.84. The null hypothesis cannot be rejected.

(C) The calculated chi-square value is 3.91, and the critical value is 5.99. The null hypothesis can be rejected.

(D) The calculated chi-square value is 3.91, and the critical value is 7.82. The null hypothesis cannot be rejected. ✓

Answer D

Correct. Because the null hypothesis is based on a model of independent assortment, the phenotypic ratios are expected to be $9{:}3{:}3{:}1$. Using the equation $\chi^2 = \Sigma\frac{(o-e)^2}{e}$, the calculated chi-square value is 3.91. With three degrees of freedom (one less than the number of phenotypic categories) and a significance level of $p = 0.05$, the critical value is 7.82. Because the calculated chi-square value is less than the critical value, the null hypothesis cannot be rejected. In other words, the data collected in the genetics experiment fit a model of independent assortment.

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R. C. Punnett conducted experiments on the inheritance of traits in the sweet pea, *Lathyrus odoratus*. In one experiment, he crossed two different true-breeding sweet pea plant strains, one with erect petals and long pollen, and the other with hooded petals and round pollen. All the offspring (F_1 generation) had erect petals and long pollen (Figure 1).

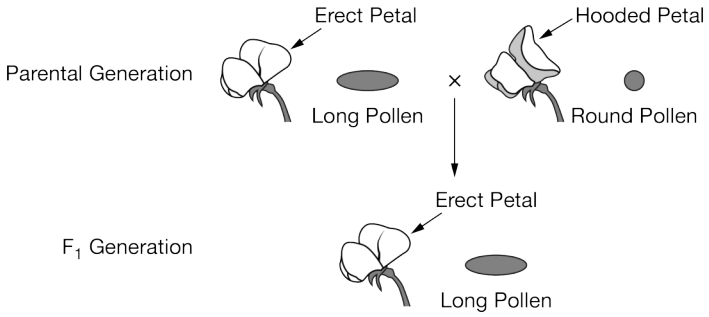


Figure 1. Sweet pea plant cross

Next, Punnett allowed the F_1 generation to self-fertilize and recorded the phenotypes of their offspring. The data are shown in Table 1.

Table 1. F_2 results of F_1 self-fertilization cross

Phenotype of F_2	Number of Offspring
Erect petals, long pollen	765
Erect petals, round pollen	62
Hooded petals, long pollen	71
Hooded petals, round pollen	220
Total	1,118

28. How many degrees of freedom should be used when looking up the critical value for a chi-square analysis of the ratios of phenotypes observed among the F_2 offspring versus the expected phenotypic ratio assuming independent assortment?

(A) 2

(B) 3 ✓

(C) 4

(D) 5

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Answer B

Correct. The number of degrees of freedom is equal to the number of categories minus 1. In this experiment, there are 4 different phenotype combinations, and $4 - 1 = 3$.

29. Sweet pea plants have a diploid $(2n)$ chromosome number of 14.

Which of the following best explains how the sweet pea plants in the parental generation produce F_1 offspring with 14 chromosomes?

(A) Meiosis I and II lead to the formation of cells with 14 chromosomes. When two cells combine during fertilization, extra chromosomes are randomly broken down, leading to offspring with 14 chromosomes.

(B) Meiosis I and II lead to the formation of cells with 14 chromosomes. When two cells combine during fertilization, extra chromosomes with recessive traits are broken down, leading to offspring with 14 chromosomes.

(C) Meiosis I and II lead to the formation of cells with 7 chromosomes. During meiosis I, homologous chromosomes separate. During meiosis II, sister chromatids separate. Two cells combine during fertilization to produce offspring with 14 chromosomes. ✓

(D) Meiosis I and II lead to the formation of cells with 7 chromosomes. During meiosis I, sister chromatids separate. During meiosis II, homologous chromosomes separate. Two cells combine during fertilization to produce offspring with 14 chromosomes.

Answer C

Correct. Meiosis produces haploid cells with half the number of chromosomes of the parent. When cells combine during fertilization, the diploid chromosome number is restored. Before meiosis, chromosomes replicate, forming duplicated chromosomes, each made of two sister chromatids. During meiosis I, homologous duplicated chromosomes line up in pairs, forming a set of 4 chromatids. The homologous chromosomes separate, forming cells with half the normal number of chromosomes, but the chromosomes are still duplicated. In meiosis II, the sister chromatids separate, forming cells with half the normal number of unduplicated chromosomes.

30. Which of the following questions would be most useful to researchers trying to determine the role of meiosis in the F_2 phenotypic frequencies?

(A) What is the molecular mechanism underlying the dominance of erect petals and long pollen?

(B) Which phenotypes give pea plants the highest level of fitness: erect or hooded petals and long or round pollen?

(C) How do the phases of meiosis differ between sweet pea plants and other organisms?

(D) What is the recombination frequency between the genes for petal shape and pollen shape? ✓

5 Heredity MCQ

Answer D

Correct. The F_2 generation in this experiment showed a greater-than-expected frequency of phenotype combinations from the parental generation. The erect petals, round pollen, and hooded petals, long pollen phenotype combinations are the result of crossing over of homologous chromosomes during meiosis I. The recombination frequency in this experiment was $\frac{(62+71)}{1118} = 12\ \%$. The closer genes are to each other, the lower the chance of them being separated from each other by crossing over and the lower the recombination frequency.

A male fruit fly (*Drosophila melanogaster*) with red eyes and long wings was mated with a female with purple eyes and vestigial wings. All of the offspring in the F_1 generation had red eyes and long wings. These F_1 flies were test crossed with purple-eyed, vestigial-winged flies. Their offspring, the F_2 generation, appeared as indicated below.

	F_2 Generation
125	red eyes, long wings
124	purple eyes, vestigial wings
18	purple eyes, long wings
16	red eyes, vestigial wings
283	Total

31. If in the F_1 and F_2 generations the same characteristics appeared in both males and females, it would be safe to assume that these traits for eye color and wing length

(A) are sex-linked

(B) vary in dominance according to sex

(C) are sex-influenced characteristics

(D) are autosomal characteristics ✓

(E) follow the Mendelian rule of independent assortment

32. In the F_2 generation, the results are best explained by the fact that

(A) the test cross with the F_1 flies resulted in sterile offspring

(B) these genes for eye color and wing shape do not pass through the F_1 generation

(C) these genes for eye color and wing shape are found on the same chromosome ✓

(D) crossing-over decreases variability

(E) the genes are sex-linked

---

5 Heredity MCQ

Dystrophin is a protein that is expressed in certain muscle cells. In combination with other cellular proteins, dystrophin strengthens protein fibers in muscle cells to allow muscles to contract without injury.

Nucleotide deletions in the gene that encodes dystrophin are associated with the genetic disorder Duchenne Muscular Dystrophy (DMD). Individuals with DMD do not produce functional dystrophin and, as a result, the protein fibers, and then entire muscle cells, become damaged.

The history of DMD for three generations of a family is shown in Figure 1.

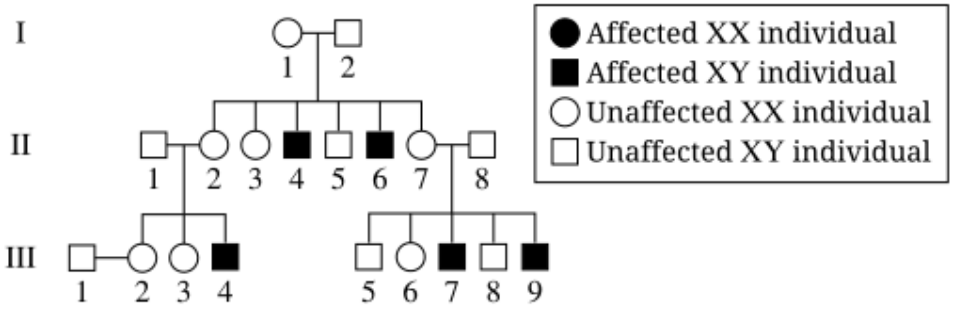


Figure 1. Pedigree showing instances of DMD in a family

Individuals III–1 and III–2 plan to have children and wish to first determine whether individual III–2 is a carrier of DMD. Individual III–2 undergoes genetic testing to determine whether individual III-2 carries a particular allele for the mutated dystrophin that is associated with the disorder in this family. The results of gel electrophoresis analysis of the individual's dystrophin alleles and the alleles of several family members are shown in Figure 2.

5 Heredity MCQ

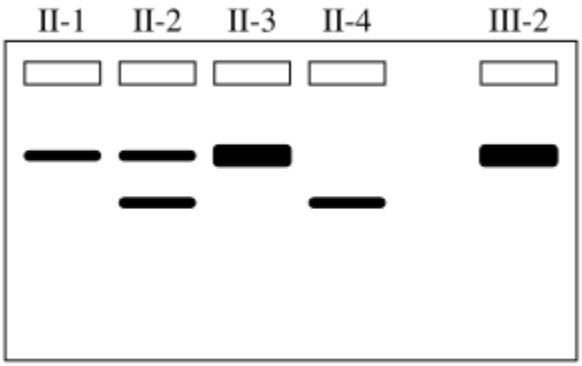


Figure 2. Gel electrophoresis showing the dystrophin alleles present in five members of the family

33. Based on the information provided, which of the following describes the most likely pattern of inheritance for the disorder?

(A) Autosomal recessive

(B) Autosomal dominant

(C) X-linked recessive ✓

(D) Mitochondrial

Answer C

Correct. The trait shows a recessive inheritance pattern because unaffected parents can produce affected offspring. The disorder is most likely X-linked because XY individuals are more likely to be affected. Based on Figure 2, XY individual $\mathrm{II}-4$ appears to have only one copy of the allele, further supporting that the allele is on the X chromosome.

34. Based on the information provided in Figures 1 and 2, which of the following best describes the chances of individuals $\mathrm{III}{-}1$ and $\mathrm{III}{-}2$ passing on the mutant dystrophin allele and/or DMD disorder to their offspring?

(A) There is a 50% chance that an XY child will have DMD.

(B) There is a 25% chance that a child will have DMD.

(C) There is a 0% chance that any of the children will have the disorder or be carriers of the mutant allele. ✓

(D) There is a 0% chance that an XY child will have DMD but a 50% chance that an XX child will be a carrier of the mutant allele.

5 Heredity MCQ

Answer C

Correct. Individual $\mathrm{III}-2$ is homozygous for the nonmutated dystrophin allele, as confirmed by the single upper band in Figure 2, and individual $\mathrm{III}-1$ does not possess the mutant allele either because this individual is XY and unaffected. If both parents do not possess the mutant allele, they will not be able to pass it on to their offspring, which means the offspring have 0% chance of having the disorder or being a carrier.

---

35. Eye color in a particular strain of fly is influenced by one gene with two alleles: a dominant allele that results in red eyes and a recessive allele that results in sepia eyes.

A red-eyed female from a true-breeding population is mated with a sepia-eyed male. The F1 offspring are all red-eyed. The F1 flies are allowed to interbreed, producing the following in the F2 generation.

Females: 40 red eyes; 13 sepia eyes

Males: 39 red eyes; 11 sepia eyes

Which of the following best describes the likely mode of inheritance for the eye-color gene?

(A) The eye-color gene is likely autosomal because males and females have similar phenotype ratios. ✓

(B) The eye-color gene is likely autosomal because more females have sepia eyes than males do.

(C) The eye-color gene is likely sex-linked because the males and females have similar phenotype ratios.

(D) The eye-color gene is likely sex-linked because the males and females display both phenotypes.

5 Heredity MCQ

36. Researchers hypothesized that red eye color in Japanese koi, a type of fish, is due to a mutation. To study the inheritance of red eye color in koi, the researchers conducted breeding experiments over several generations. The results are summarized in Figure 1.

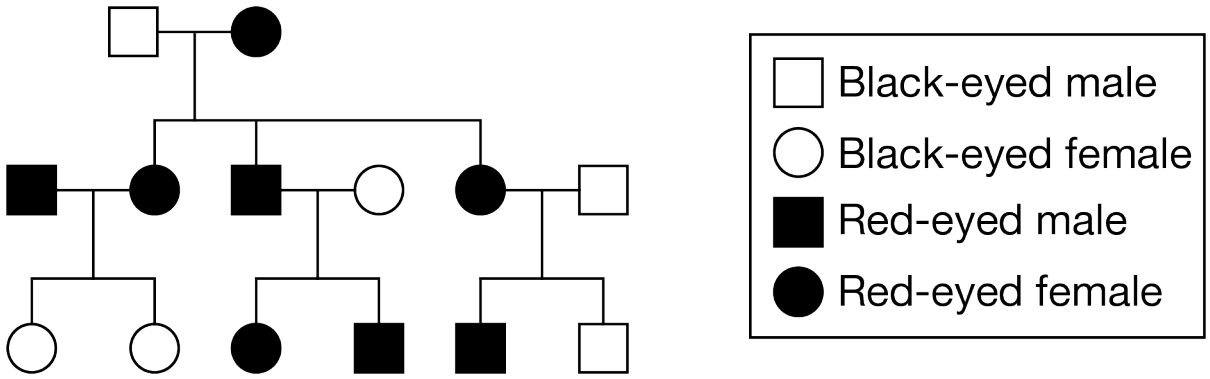


Figure 1. Inheritance of eye color in Japanese koi

Based on the data in Figure 1, which of the following is the best prediction of the mode of inheritance of red eyes in Japanese koi?

(A) The allele for red eyes is inherited in an autosomal dominant pattern. ✓

(B) The allele for red eyes is inherited in an autosomal recessive pattern.

(C) The allele for red eyes is inherited in an X-linked recessive pattern.

(D) The allele for red eyes is inherited in an X-linked dominant pattern.

Answer A

Correct. Based on the pedigree, red eye color in koi is inherited in an autosomal dominant pattern. In the second generation, two koi with red eyes produced female offspring with black eyes. This could only happen if the parents were heterozygous with the dominant allele.

37. In sheep, eye color is controlled by a single gene with two alleles. When a homozygous brown-eyed sheep is crossed with a homozygous green-eyed sheep, blue-eyed offspring are produced. If the blue-eyed sheep are mated with each other, what percent of their offspring will most likely have brown eyes?

5 Heredity MCQ

(A) 0%

(B) 25% ✓

(C) 50%

(D) 75%

(E) 100%

38. Eye pigment in a particular strain of fly is determined by two genes. An autosomal gene that controls the color of the pigments in the eye has two alleles: a dominant allele *(*R*)* that results in red eyes and a recessive allele *(r)* that results in sepia eyes. A sex-linked gene that controls the expression of the colored pigments also has two alleles: a dominant allele *(*T*)* that allows for expression of the colored pigments and a recessive allele *(t)* that does not allow for expression of the colored pigments. Individuals without a T allele have white eyes regardless of the alleles of other eye-color genes.

Which of the following represents a cross between a white-eyed female and a red-eyed male?

(A) $Tt\mathrm{X^RX^R} \times tt\mathrm{X^rY}$

(B) $Tt\mathrm{X^rX^r} \times tt\mathrm{X^RY}$

(C) $RR\mathrm{X}^T\mathrm{X}^T \times Rr\mathrm{X}^T\mathrm{Y}$

(D) $Rr\mathrm{X}^t\mathrm{X}^t \times Rr\mathrm{X}^T\mathrm{Y}$ ✓

5 Heredity MCQ

In a laboratory population of fruit flies, body color is determined by a single gene with a wild-type allele associated with black body color and a mutant allele associated with yellow body color. A group of researchers hypothesizes that the mutation in fruit flies that causes the yellow body is sex-linked. They also hypothesize that this mutation results in a change in the structure of the sex combs, bristle-like structures on the front legs of male flies that facilitate mating.

To determine the inheritance pattern of the mutant allele, researchers crossed yellow-bodied females with wild-type males. Flies from the F_1 generation were then crossed. The distribution of phenotypes observed in the F_1 and F_2 generations is shown in Table 1.

TABLE 1. DISTRIBUTION OF PHENOTYPES OBSERVED IN THE P, F_1, and F_2 GENERATIONS

	Wild-Type Females*	Yellow-Bodied Females*	Wild-Type Males**	Yellow-Bodied Males**
P Generation	0	3	3	0
F_1 Generation	38	0	0	41
F_2 Generation	117	124	121	130

*Female fruit flies have two X chromosomes.

**Male fruit flies have one X chromosome and one Y chromosome.

39. Based on the researchers' hypothesis that the mutant allele is sex-linked, which of the following is closest to the expected number of wild-type females and the expected number of yellow-bodied males in the F_2 generation?

(A) 123 wild-type females and 123 yellow-bodied males ✓

(B) 61 wild-type females and 184 yellow-bodied males

(C) 123 wild-type females and 369 yellow-bodied males

(D) 184 wild-type females and 123 yellow-bodied males

Answer A

Correct. Based on Mendel's Laws of Segregation and Independent Assortment, and based on the P generation, the phenotypic ratio of the F_2 generation should be 1 : 1 : 1 : 1. Thus, 123 wild-type females and 123 yellow-bodied males would most closely match the expected numbers.

5 Heredity MCQ

40. Using a p-value of 0.05, researchers performed a chi-square analysis of the distribution of phenotypes in the F_2 generation.

Which of the following chi-square calculated values would cause the researchers to reject their null hypothesis?

(A) 0.73

(B) 2.16

(C) 4.35

(D) 7.98 ✓

Answer D

Correct. Because four different phenotypes would be produced in the F_2 generation, there should be 3 degrees of freedom in the chi-square analysis. Based on the chi-square table, a chi-square value that is greater than 7.81 is cause to reject the null hypothesis. The 7.98 value is higher than 7.81, and thus the researchers should reject the null hypothesis.

41. Based on the data in Table 1, which of the following is the most likely inheritance pattern for the mutant allele?

(A) The mutant allele is X-linked dominant.

(B) The mutant allele is X-linked recessive. ✓

(C) The mutant allele is autosomal (non-sex-linked) recessive.

(D) The mutant allele is Y-linked.

Answer B

Correct. To produce the values in the table, the mutant allele must be an X-linked recessive trait. Thus, in the P generation, if X^B represents the allele for black bodies and X^b represents the allele for yellow bodies, the 3 females would be X^bX^b and the 3 males would be X^By. In the F_1 generation, all females would be X^BX^b, and all males would be X^bY. The F_2 generation should produce an approximately $1:1:1:1$ ratio of X^BX^b, X^bX^b, X^By, X^bY flies.

42. Which of the following most likely explains how the sex combs of yellow-bodied flies have a different structure than do the sex combs of wild-type flies?

5 Heredity MCQ

(A) The protein encoded by the mutant body color allele interferes with the development of sex combs. ✓

(B) The protein encoded by the mutant body color allele is only produced if a copy of the wild-type body color allele is present.

(C) The gene associated with body color and the gene associated with sex comb development both exhibit codominance in flies.

(D) The gene associated with body color and the gene associated with sex comb development are located far apart on the same chromosome in flies.

Answer A

Correct. It is likely that the protein encoded by the mutant body color allele is also important to the normal development of sex combs in male flies.

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43. The relative location of four genes on a chromosome can be mapped from the following data on crossover frequencies

Genes	*Frequency of Crossover*
B and *D*	5%
C and *A*	15%
A and *B*	30%
C and *B*	45%
C and *D*	50%

Which of the following represents the relative positions of these four genes on the chromosome?

(A) *ABCD*

(B) *ADCB*

(C) *CABD* ✓

(D) *CBAD*

(E) *DBCA*

44. Which of the following occurs in all species of living organisms and may lead to an increase in genetic variation?

(A) Mutations in the genome ✓

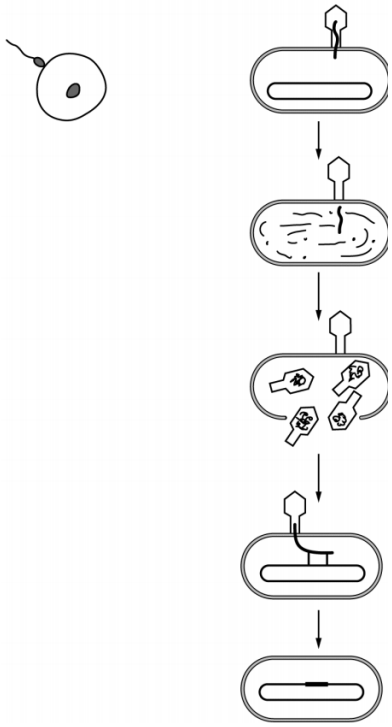
(B) Crossing-over in meiosis

(C) Random assortment of chromosomes

(D) Alternative splicing of mRNA

5 Heredity MCQ

45.



Which of the following correctly compares the two processes shown above?

(A) Both processes increase genetic variation. ✓

(B) Both processes represent aspects of sexual reproduction.

(C) Both processes require partitioning of genetic material and organelles.

(D) The amount of genetic material per cell remains constant in both processes.

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days, and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

	Dish A		Dish B	
	Day 7 Covered	Day 14 Uncovered	Day 7 Uncovered	Day 14 Uncovered
Germinated seeds	12	20	20	20
Green-leaved seedlings	0	14	15	15
Yellow-leaved seedlings	12	6	5	5
Mean stem length below first set of leaves	8 mm	9 mm	3 mm	3 mm

5 Heredity MCQ

46. Which of the following best supports the hypothesis that the difference in leaf color is genetically controlled?

(A) The number of yellow-leaved seedlings in dish A on day 7

(B) The number of germinated seeds in dish A on days 7 and 14

(C) The death of all the yellow-leaved seedlings

(D) The existence of yellow-leaved seedlings as well as green-leaved ones on day 14 in dish B ✓

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

	Dish A		Dish B	
	Day 7 Covered	Day 14 Uncovered	Day 7 Uncovered	Day 14 Uncovered
Germinated seeds	12	20	20	20
Green-leaved seedlings	0	14	15	15
Yellow-leaved seedlings	12	6	5	5
Mean stem length below first set of leaves	8 mm	9 mm	3 mm	3 mm

47. Which of the following best supports the hypothesis that the difference in leaf color is genetically controlled?

(A) The number of yellow-leaved seedlings in dish A on day 7

(B) The number of germinated seeds in dish A on days 7 and 14

(C) The death of all the yellow-leaved seedlings

(D) The existence of yellow-leaved seedlings as well as green-leaved ones on day 14 in dish B ✓

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48. Gregor Mendel's pioneering genetic experiments with pea plants occurred before the discovery of the structure and function of chromosomes. Which of the following observations about inheritance in pea plants could be explained only after the discovery that genes may be linked on a chromosome?

(A) Pea color and pea shape display independent inheritance patterns.

(B) Offspring of a given cross show all possible combinations of traits in equal proportions.

(C) Most offspring of a given cross have a combination of traits that is identical to that of either one parent or the other. ✓

(D) Recessive phenotypes can skip a generation, showing up only in the parental and F_2 generations.

5 Heredity MCQ

49. Sex chromosomes determine the phenotype of sex in humans. Embryos containing XX chromosomes develop into females, and embryos containing XY chromosomes develop into males. The sex chromosomes separate during meiosis, going to different gamete cells.

A woman is heterozygous for the X-linked recessive trait of hemophilia A. Her sex chromosomes can be designated as X^HX^h. During meiosis the chromosomes separate as shown in Figure 1.

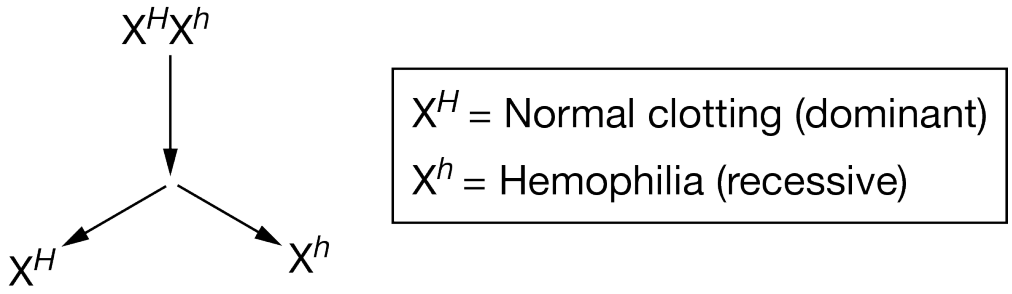


Figure 1. Transmission pattern for sex chromosomes of a woman heterozygous for hemophilia A into gametes

If the woman and a man with normal clotting function have children, what is the probability of their children exhibiting hemophilia A?

(A) 50 percent for daughters, 0 percent for sons

(B) 50 percent for sons, 0 percent for daughters ✓

(C) 50 percent for all children

(D) 0 percent for all children

Answer B

Correct. Half of the sons would inherit the X^H allele from the woman and a Y chromosome from the man. Without a X^H allele, these sons would develop hemophilia A.

50. Hydrangea flowers differ in color based on interactions between a pigment in the flower cells and aluminum ions dissolved in water absorbed from the soil. When aluminum ions are present, the pigment is blue; when aluminum ions are absent, the pigment is pink. The amount of aluminum ions the plant can absorb is dependent on the pH of the soil.

Which of the following questions will best help a researcher design an experiment to learn about the relationship between hydrangea flower color, aluminum, and pH?

5 Heredity MCQ

(A) How do aluminum-pigment interactions change the structure of the pigment and flower color?

(B) Can the pigment be used as an indicator of pH, because flower color depends on pH and aluminum?

(C) Does an increase in hydrogen ion concentration affect the ability of the pigment to dissolve in water?

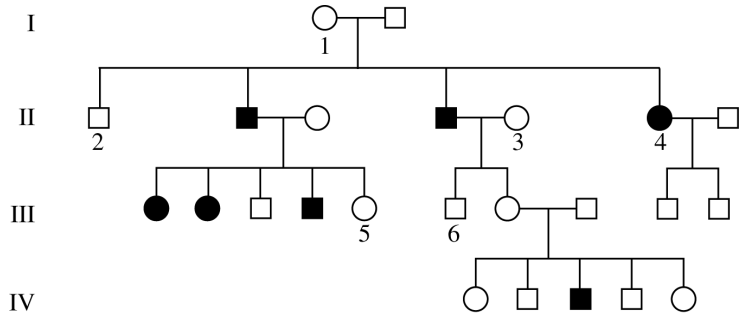
(D) What will be the flower color of hydrangeas grown in soils of varying pH with or without aluminum? ✓

Answer D

Correct. This question addresses how aluminum and pH interact to affect flower color.

For following group of questions first study the description of the situation and diagram and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

The pedigree of a family with a history of a particular genetic disease is shown below. Squares represent males and circles represent females. Shaded symbols represent those who have the disease.



51. If Individual 2 were to marry a woman with no family history of the disease, which of the following would most likely be true of their children?

(A) All of the children would have the disease.

(B) None of the children would have the disease. ✓

(C) Only the sons would have the disease.

(D) All of the sons would be carriers of the disease.

(E) None of the daughters would be carriers of the disease.

52. If Individual 6 marries a woman with the disease, what is the probability that their first child will have the disease?

5 Heredity MCQ

(A) 0

(B) 25%

(C) 50% ✓

(D) 75%

(E) 100%

53. Which of the following is the most likely mode of inheritance for the disease?

(A) Codominance

(B) Autosomal dominant

(C) Autosomal recessive ✓

(D) Sex-linked dominant

(E) Sex-linked recessive

---

54. Individuals with an inherited autosomal recessive disorder called primary ciliary dyskinesia (PCD) often have severe respiratory problems due to defective cilia. Males with PCD are often sterile because they produce sperm with defective flagella. Which of the following most likely explains the effect of the recessive allele?

(A) The mitochondria are defective and do not produce sufficient protein to synthesize microtubules in the cilia and flagella.

(B) The plasma membrane of the alveoli is not permeable to carbon dioxide during respiration because it is too hydrophobic.

(C) The Golgi bodies secrete an enzyme that destroys the proteins in the flagella and cilia.

(D) The cells do not produce functional motor proteins in flagella and cilia. ✓

55. In garden peas, a single gene controls stem length. The recessive allele (*t*) produces short stems when homozygous. The dominant allele (*T*) produces long stems. A short-stemmed plant is crossed with a heterozygous long-stemmed plant. Which of the following represents the expected phenotypes of the offspring and the ratio in which they will occur?

(A) 3 long-stemmed plants: 1 short-stemmed plant

(B) 1 long-stemmed plant: 1 short-stemmed plant ✓

(C) 1 long-stemmed plant: 3 short-stemmed plants

(D) Long-stemmed plants only

(E) Short-stemmed plants only

5 Heredity MCQ

56. The Russian blue is a rare breed of cat that is susceptible to developing cataracts on the eyes. Scientists hypothesize that this condition is inherited as a result of a mutation. Figure 1 shows a pedigree obtained in a study of cats owned by members of the Russian Blue Club in Sweden.

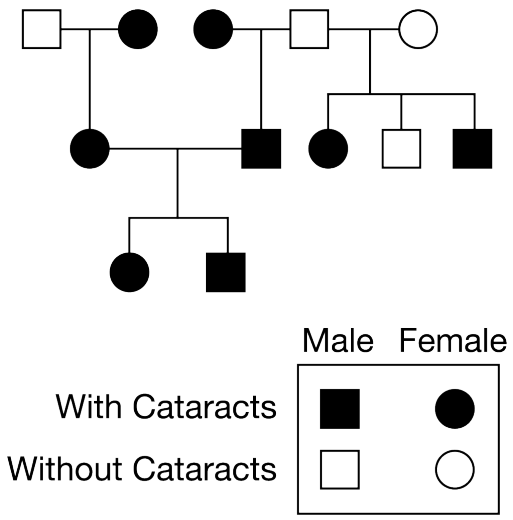


Figure 1. Pedigree of Russian blue cats owned by Russian Blue Club

Based on the inheritance pattern shown in Figure 1, which of the following best predicts the nature of the original mutation?

(A) A recessive mutation on the X chromosome

(B) A recessive mutation on a somatic chromosome ✓

(C) A dominant mutation on the X chromosome

(D) A dominant mutation on a somatic chromosome

Answer B

Correct. The inheritance pattern shows offspring phenotypes that are possible only if the trait is somatic and recessive. Since two cats without cataracts produce cats with cataracts, the trait must be recessive. This trait cannot be X-linked recessive because unaffected males are able to pass on the trait to their offspring.

57. In humans, red-green color blindness is a sex-linked recessive trait. If a man and a woman produce a color-blind son, which of the following must be true?

5 Heredity MCQ

(A) The father is color-blind.

(B) Both parents carry the allele for color blindness.

(C) Neither parent carries the allele for color blindness.

(D) The father carries the allele for color blindness.

(E) The mother carries the allele for color blindness. ✓

58. In peas the trait for tall plants is dominant (*T*) and the trait for short plants is recessive (*t*). The trait for yellow seed color is dominant (*Y*) and the trait for green seed color is recessive (*y*). A cross between two plants results in 296 tall yellow plants and 104 tall green plants. Which of the following are most likely to be the genotypes of the parents?

(A) *TTYY* x *TTYY*

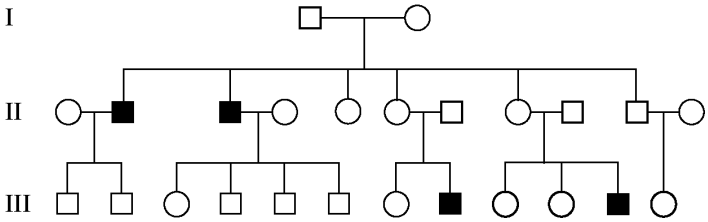
(B) *TTyy* x *TTYy*

(C) *TtYy* x *TtYy*

(D) *TtYy* x *TTYy* ✓

(E) *TtYY* x *Ttyy*

59. In the following human pedigree, squares represent XY individuals, circles represent XX individuals, and shaded symbols indicate individuals affected with a disorder.



One of the affected XY individuals from the third generation has a child with an XX individual who is a carrier. For the pedigree shown above, which of the following best expresses the probability that the couple's first XY offspring will be affected with the disorder?

(A) 25%

(B) 50% ✓

(C) 75%

(D) 100%

5 Heredity MCQ

60.

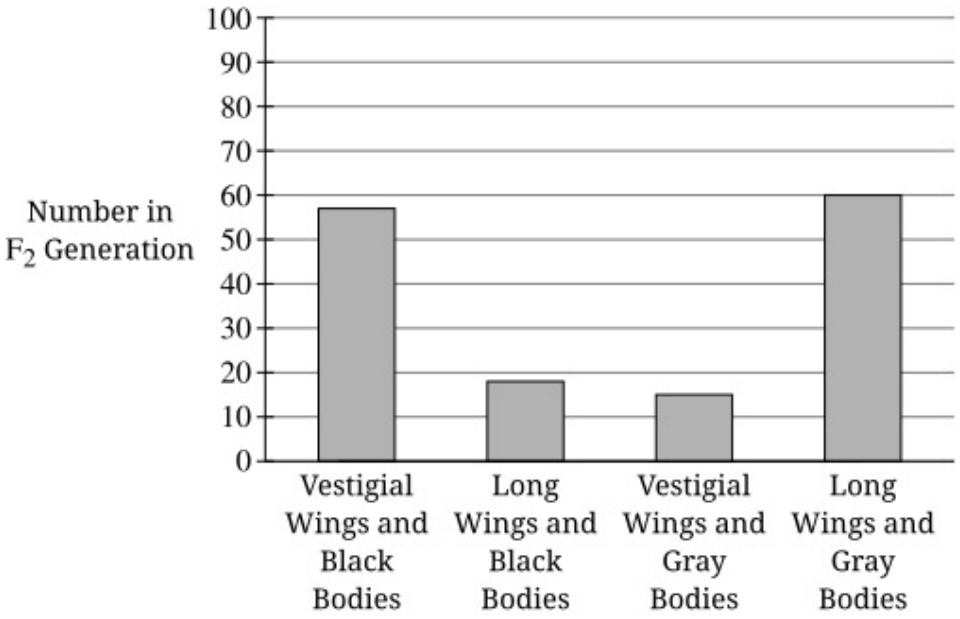


Figure 1. The distribution of phenotypes in the F_2 generation of fruit flies

In fruit flies, genes that control body color and wing length are linked. A researcher crossed a true-breeding fruit fly with long wings and a gray body with a true-breeding fruit fly with vestigial wings and a black body. All of the F_1 flies had long wings and gray bodies. The researcher then crossed the F_1 flies with true-breeding flies that have vestigial wings and black bodies. The number of flies with each phenotype in the resulting F_2 generation is shown in Figure 1.

What percent of the F_2 generation flies carry chromosomes produced by crossing-over between the two linked genes?

(A) 10%

(B) 15%

(C) 22% ✓

(D) 33%

Answer C

Correct. There are 33 flies with crossing-over phenotypes (18 have long wings with black bodies, and 15 have vestigial wings with gray bodies). The total number of 33 is then divided by 150, the total number of F_2 flies: $\frac{33}{150} = 22\%$.

5 Heredity MCQ

61. In sweet pea plants, purple flower color is dominant over red flower color and long pollen grain shape is dominant over round pollen grain shape. Two sweet pea plants that are heterozygous for both flower color and pollen grain shape are crossed with one another. A geneticist is surprised to observe that there are far fewer round, purple offspring and long, red offspring than were predicted by the $9:3:3:1$ expected phenotypic ratio.

Which of the following statements provides the most reasonable prediction to account for the deviation from the expected results?

(A) In sweet pea plants, the gene for flower color and the gene for pollen grain shape are genetically linked. ✓

(B) In sweet pea plants, the genes for flower color and for pollen grain shape both exhibit codominance.

(C) Several mutations occurred spontaneously producing a deviation from the expected phenotypic ratios of the offspring.

(D) The genes for flower color and pollen grain shape are inherited independently because of the law of independent assortment.

Answer A

Correct. There are fewer recombinant phenotypes than expected due to genetic linkage. The gene for flower color and the gene for pollen grain shape have a strong tendency to segregate as a unit, so it is most likely that they are close together on the same chromosome and thus, the offspring exhibit parental phenotypes more than expected.

62. Four trials measuring recombination frequency between gene R and gene L were conducted, and the results are shown in Table 1.

Table 1. Recombination frequency

	Trial 1	Trial 2	Trial 3	Trial 4
Frequency of recombination	24%	33%	26%	29%

The mean map distance between gene R and gene L is closest to which of the following?

(A) 0.28 map units

(B) 28 map units ✓

(C) 0.14 map units

(D) 14 map units

5 Heredity MCQ

Answer B

Correct. The recombination frequency is equal to the map distance between genes on the same chromosome. Thus, the mean recombination frequency should be determined, and this is equal to the mean map distance between the two genes.

63. Two fruit fly mutations are ebony body (e) and sepia eyes (s). Four different students performed dihybrid crosses with flies that were heterozygous with a mutant allele and a wild-type allele for both genes $(EeSs \times EeSs)$. The results are shown in Table 1.

Table 1. Offspring of four separate dihybrid crosses

	Number of Offspring			
Phenotype	**Student 1**	**Student 2**	**Student 3**	**Student 4**
Wild-type eyes, wild-type body	89	92	83	95
Wild-type eyes, ebony body	22	17	15	20
Sepia eyes, wild-type body	21	24	30	16
Sepia eyes, ebony body	28	27	32	29

The mean number of fruit flies per student that are homozygous recessive for both genes is closest to which of the following?

(A) 89.75

(B) 29.0 ✓

(C) 22.75

(D) 18.5

Answer B

Correct. The mean number of flies that express both recessive mutations homozygous recessive is 29.0. The sepia eyes, ebony body phenotype is homozygous recessive $(eess)$.

5 Heredity MCQ

64. A model showing the cells in anaphase I and anaphase II of meiosis during a nondisjunction event is shown in Figure 1.

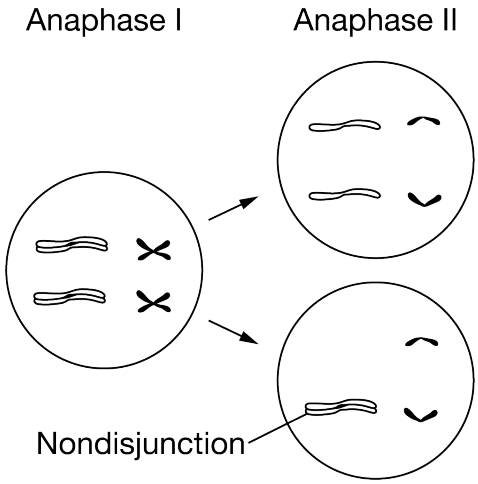


Figure 1. Model of a nondisjunction event

Which of the following best predicts the effect of the chromosomal segregation error shown in Figure 1?

(A) All of the resulting gametes will have an extra chromosome.

(B) All of the resulting gametes will be missing a chromosome.

(C) Half of the resulting gametes will have an extra chromosome and the other half will be missing a chromosome.

(D) Half of the resulting gametes will have the correct number of chromosomes, and the other half will have an incorrect number of chromosomes. ✓

Answer D

Correct. In this instance, nondisjunction of sister chromatids during meiosis II will produce two gametes with the correct number of chromosomes (n). The remaining two gametes that are produced will be $n + 1$ and $n - 1$, respectively.

5 Heredity MCQ

Mice live in territorial social groups, and each group has a single dominant male with the rest of the males in the group being subordinate. Male mice mark the area in which they live with their scent by leaving urine wherever they walk. There are two classes of proteins that are found in mouse urine, major urinary proteins (MUPs) and major histocompatibility complex (MHC) proteins. The proteins provide a unique scent for each mouse.

In an experiment, wild male mice were kept isolated in separate cages and their urination patterns were recorded. They were then combined into a single cage that was divided into two compartments by a mesh screen, and again their urination patterns were recorded (Figure 1).

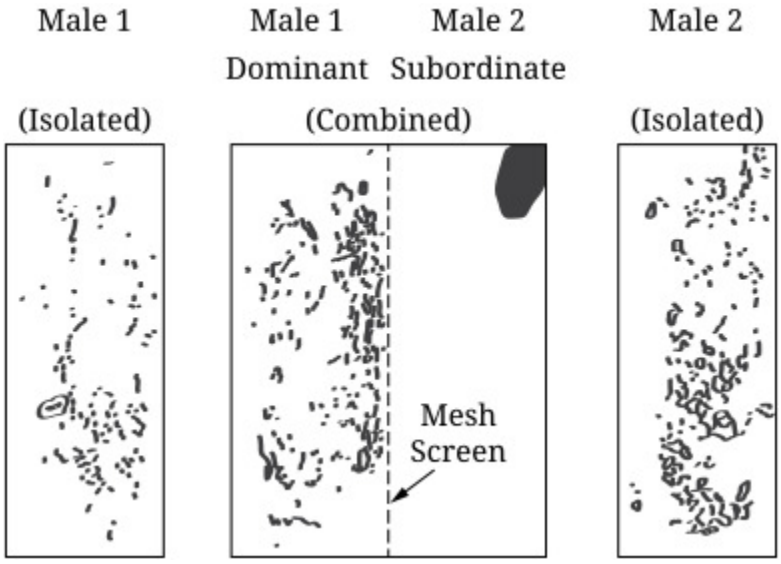


Figure 1. Urination patterns recorded during an experiment

65. Which of the following best explains why particular MUP protein variants are associated with male dominance in individuals in populations of wild mice?

(A) Viral infections introduced new MUP alleles into affected populations.

(B) Environmental stress caused a decreased rate of mutations in the MUP genes of subordinate mice.

(C) Interbreeding between different mouse species introduced new MUP alleles into the populations.

(D) The MUP genes are located on a chromosome near genes associated with phenotypes of dominant males. ✓

Answer D

Correct. Male dominance is an inherited trait encoded by particular alleles of genes. Genes that are located near each other on the same chromosome tend to stay together during meiosis. This explains why certain MUP variants are associated with other traits that produce dominant males.

5 Heredity MCQ

66. In order to determine the effects of age on the accumulation of mitochondrial mutations, mitochondrial DNA samples from young mice (3 months) and old mice (30 months) were observed for mutations.

Table 1. Number of mitochondrial mutations in 10^6 base pairs

Mouse	Young	Old
A	1	
B	1	
C	3	
D	2	
E	4	
F		10
G		15
H		12
I		9
J		11

Which of the following is a correct analysis of this data set?

(A) There is an increase in the mean number of mutations for the two age groups of 9.0 mutations per 10^6 base pairs. This is more critical in male mammals since mitochondria are paternally inherited.

(B) There is an increase in the mean number of mutations for the two age groups of 9.0 mutations per 10^6 base pairs. This is more critical in female mammals since mitochondria are maternally inherited.

(C) There is an increase in the mean number of mutations for the two age groups of 9.2 mutations per 10^6 base pairs. This is more critical in male mammals since mitochondria are paternally inherited.

(D) There is an increase in the mean number of mutations for the two age groups of 9.2 mutations per 10^6 base pairs. This is more critical in female mammals since mitochondria are maternally inherited. ✓

Answer D

Correct. The mean number of mutations for the young mice is $\frac{1+1+3+2+4}{5} = 2.2$ per 10^6 base pairs, and the mean number of mutations for the old mice is $\frac{10+15+12+9+11}{5} = 11.4$ per 10^6 base pairs. This yields an increase in the mean number of mutations of 9.2 per 10^6 base pairs between the young and old mice. Mitochondria are normally passed to offspring only from their mother.

5 Heredity MCQ

67. A true-breeding variety of wheat that produces deep-red-colored grain was crossed with a true-breeding variety that produces a white-colored grain. The resulting F_1 offspring all had medium-red-colored grain. Five crosses of the F_1 plants were all grown under the same conditions and resulted in plants with a variety of grain color, as indicated in Table 1.

Table 1. F_2 phenotypes resulting from five crosses

Grain Color	Cross 1	Cross 2	Cross 3	Cross 4	Cross 5
Deep red	1	2	1	2	0
Dark red	6	7	7	5	6
Red	15	13	16	17	14
Medium red	20	22	21	20	21
Light red	15	16	14	16	15
Very light red	6	4	3	3	6
White	1	0	2	1	2

Which of the follow indicates the mean number per cross of F_2 plants producing medium-red grain and correctly explains the distribution of the phenotypes?

(A) The mean number of medium-red phenotypes per cross is 20.8. The distribution of phenotypes suggests that grain color is under environmental control.

(B) The mean number of medium-red phenotypes per cross is 20.8. The distribution of phenotypes suggests that multiple genes are involved in grain color determination. ✓

(C) The mean number of medium-red phenotypes per cross is 104. The distribution of phenotypes suggests that grain color is under environmental control.

(D) The mean number of medium-red phenotypes per cross is 104. The distribution of phenotypes suggests that multiple genes are involved in grain color determination.

Answer B

Correct. This is the correct mean, $\frac{(20+22+21+20+21)}{5} = 20.8$. The phenotype distribution shows a continuous range of grain color from plants grown under the same conditions, suggesting that grain color is controlled by multiple alleles.

5 Heredity MCQ

68. An experiment was performed to determine the mode of inheritance of two mouse genes, one for fur color and one for fur length. It is known that black fur (B) is dominant over white fur (b) and that long fur (L) is dominant over short fur (l). To determine how the genes are inherited, a cross was performed between two true-breeding mice, one with long black fur and one with short white fur. Their progeny, the F_1 generation, all had long black fur. Five F_1 male-female pairs were then crossed with one another. The F_2 generation phenotypes for each cross are shown in Table 1.

Table 1. Number of F_2 generation phenotypes for five crosses

Phenotype	Cross 1	Cross 2	Cross 3	Cross 4	Cross 5
Long black fur	6	5	5	6	7
Long white fur	1	1	0	2	2
Short black fur	0	2	1	1	1
Short white fur	2	3	3	2	2

Which of the following is the mean number per cross of F_2 generation offspring that are the result of crossing over?

(A) 1

(B) 2.2 ✓

(C) 2.4

(D) 5.8

Answer B

Correct. This is the mean of the six F_2 offspring with long white fur and the five F_2 offspring with short black fur. These two phenotypes are both the result of crossing over. The 11 offspring divided by the 5 crosses results in a mean per cross of 2.2.

5 Heredity MCQ

69. In pea plants, purple flower color is dominant to red flower color and long pollen grains are dominant to round pollen grains. Researchers crossed two pure-breeding lines of the pea plants to investigate whether the genes controlling flower color and pollen shape segregate independently. The procedure for the genetics experiment is summarized in Figure 1.

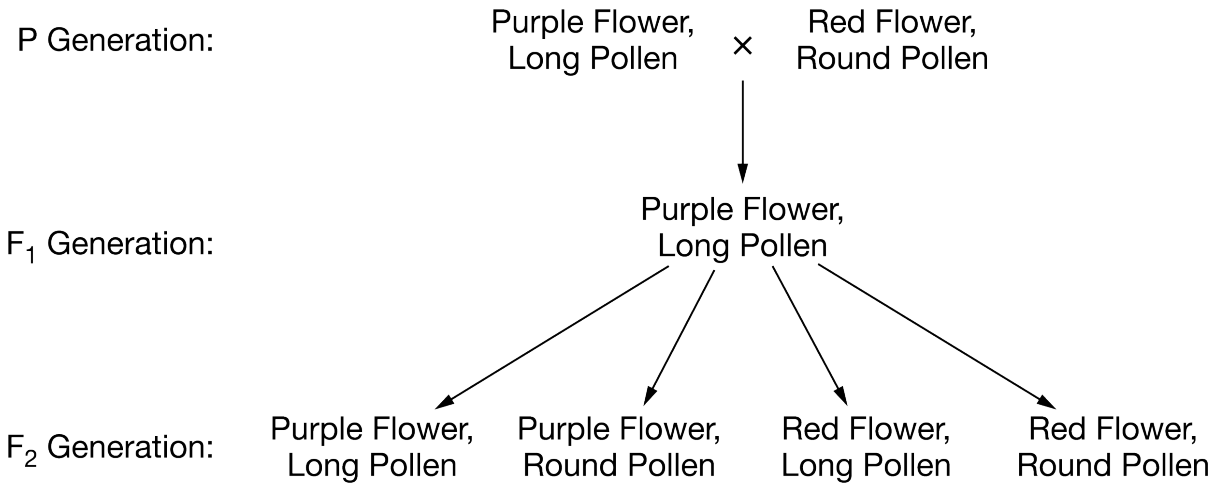


Figure 1. Summary of a genetics experiment using pea plants

Which of the following tables best shows the expected values in the F_2 generation for a chi-square goodness-of-fit test for a model of independent assortment?

5 Heredity MCQ

(A)

Phenotype	Expected
Purple, long	533
Purple, round	533
Red, long	533
Red, round	533

(B) ✓

Phenotype	Expected
Purple, long	1199
Purple, round	400
Red, long	400
Red, round	133

(C)

Phenotype	Expected
Purple, long	1067
Purple, round	355
Red, long	355
Red, round	355

(D)

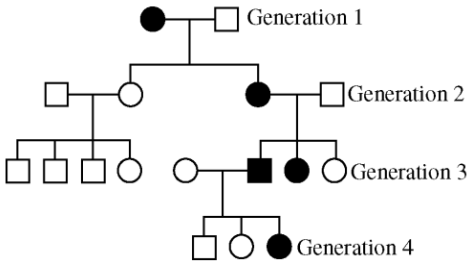
5 Heredity MCQ

Phenotype	Expected
Purple, long	1002
Purple, round	515
Red, long	515
Red, round	1002

Answer B

Correct. Because the F_1 offspring used to produce the F_2 generation were heterozygous for both traits, the expected phenotypic ratio for a model of independent assortment is $9:3:3:1$.

70. The pedigree below shows the inheritance of a dominant allele of a gene in a family over several generations. Circles represent females and squares represent males. Shaded symbols indicate individuals carrying the allele.



The pedigree suggests that the gene is on a nuclear chromosome, and not on mitochondrial DNA, because

(A) mitochondrial genes are not heritable

(B) mitochondrial mutations cannot produce dominant traits

(C) maternal mitochondrial mutations are inherited by all of a mother's offspring ✓

(D) mitochondrial DNA is circular, whereas chromosomal DNAis linear

5 Heredity MCQ

71. Pigeons demonstrate ZW sex determination, such that a ZZ genotype produces a male and a ZW genotype produces a female. The gene for feather color is located on the Z chromosome, and the red allele is dominant over the brown allele. Three crosses between brown male pigeons and red female pigeons were performed, and the results are shown below.

Table 1. Offspring from three separate crosses of a brown male pigeon and a red female pigeon

	Number of Offspring		
Phenotype	Cross 1	Cross 2	Cross 3
Red	11	9	7
Brown	9	11	13

Which of the following is the mean number of male offspring produced by the three crosses?

(A) 27

(B) 20

(C) 11

(D) 9 ✓

Answer D

Correct. For a pigeon to be red, a dominant allele must have been inherited from the female parent. Thus, red pigeons produced by the cross described above will always have two Z chromosomes, which makes them male. The genotype of the offspring would be either Z^RZ^r (phenotype: red male) or Z^rW (phenotype: brown female).

5 Heredity MCQ

72. A student crosses a pure-breeding line of red-flowered poinsettias with a pure-breeding line of white-flowered poinsettias. The student observes that all the plants in the F_1 generation have pink flowers. The student then crosses the F_1 plants with one another and records observations about the plants in the F_2 generation. The student will use the F_2 data to perform a chi-square goodness-of-fit test for a model of incomplete dominance. The setup for the student's chi-square goodness-of-fit test is presented in Table 1.

Table 1. A chi-square goodness-of-fit test for incomplete dominance

Phenotype	Observed	Expected
Red flowers	64	56
Pink flowers	107	112
White flowers	53	56

The critical value for a chi-square test with a significance level of $p = 0.05$ and 2 degrees of freedom is 5.99. Which of the following statements best completes the student's chi-square goodness-of-fit test?

(A) The calculated chi-square value is 1.53, and the null hypothesis cannot be rejected. ✓

(B) The calculated chi-square value is 1.53, and the null hypothesis can be rejected.

(C) The calculated chi-square value is 98, and the null hypothesis cannot be rejected.

(D) The calculated chi-square value is 98, and the null hypothesis can be rejected.

Answer A

Correct. The calculated chi-square value is 1.53. Because the calculated chi-square value is less than the critical value, the null hypothesis cannot be rejected. Consequently, the student may conclude that the data fit a model of incomplete dominance.

Directions: This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

This group of questions refers to the probabilities below.

Assume that the alleles referred to all assort independently.

(A) 0
(B) 1/16
(C) 1/4
(D) 1/2
(E) 3/4

5 Heredity MCQ

73. Probability that the genotype *Aa* will be produced by the parents *Aa* x *Aa*

(A) 0
(B) 1/16
(C) 1/4
(D) 1/2 ✓
(E) 3/4

74. Probability that the genotype *ccdd* will be produced by the parents *CcDd* x *CcDd*

(A) 0
(B) 1/16 ✓
(C) 1/4
(D) 1/2
(E) 3/4

75. Probability that the genotype *Rr* will be produced by the parents *Rr* x *rr*

(A) 0
(B) 1/16
(C) 1/4
(D) 1/2 ✓
(E) 3/4

76. Probability that the genotype *TTSs* will be produced by the parents *TTSs* x *TtSS*

(A) 0
(B) 1/16
(C) 1/4 ✓
(D) 1/2
(E) 3/4

5 Heredity MCQ

77. In fruit flies, purple eyes and ebony body are traits that display autosomal recessive patterns of inheritance. In a genetics experiment, students cross wild-type flies with flies that have purple eyes and ebony bodies. The students observe that all the flies in the F_1 generation have normal eyes and a normal body color. The students then allow the F_1 flies to mate and produce an F_2 generation. The students record observations about the flies in the F_2 generation and use the data to perform a chi-square goodness-of-fit test for a model of independent assortment. The setup for the students' chi-square goodness-of-fit test is presented in Table 1.

Table 1. The students' chi-square goodness-of-fit test for a model of independent assortment

Phenotype	Observed	Expected
Normal eyes, normal body	187	171
Normal eyes, ebony body	49	57
Purple eyes, normal body	41	57
Purple eyes, ebony body	27	19

The students choose a significance level of $p = 0.01$. Which of the following statements best completes the next step of the chi-square goodness-of-fit test?

(A) The calculated chi-square value is 2.11, and the critical value is 7.82.

(B) The calculated chi-square value is 2.11, and the critical value is 11.35.

(C) The calculated chi-square value is 10.48, and the critical value is 7.82.

(D) The calculated chi-square value is 10.48, and the critical value is 11.35. ✓

Answer D

Correct. The calculated chi-square value is 10.48, and the critical value for 3 degrees of freedom and a significance level of $p = 0.01$ is 11.35. Because the calculated chi-square value is less than the critical value, the null hypothesis cannot be rejected. Consequently, the students can conclude that the data fit a model of independent assortment.

5 Heredity MCQ

78.

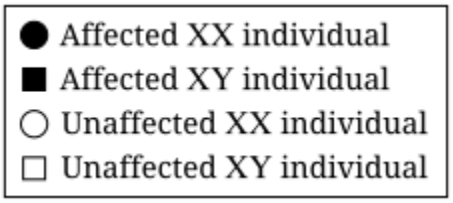
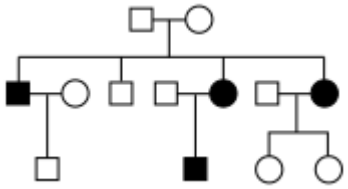
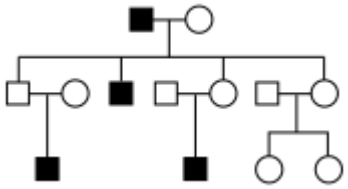
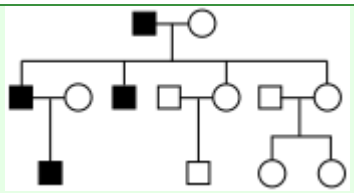


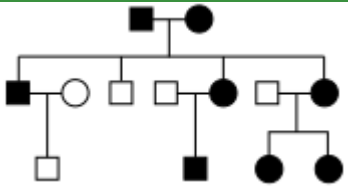
Figure 1. Key for interpreting a pedigree

Which of the following pedigrees represents the inheritance of a trait most likely associated with a gene on the Y chromosome?

(A) 

(B) 

(C)  ✓

(D) 

Answer C

Correct. A Y chromosome is only present in an XY individual and is passed by the XY individual to all XY offspring. In this pedigree, only the F_1 XY individuals inherit the trait; F_1 XX individuals do not. The only F_2 individual to inherit the trait is also an XY offspring of an F_1 individual with the trait.

5 Heredity MCQ

79. In fruit flies, sepia eyes and ebony body are traits that display autosomal recessive patterns of inheritance. To investigate whether the traits are genetically linked, students cross wild-type flies with a line of flies that have sepia eyes and ebony bodies. The students observe that all the flies in the F_1 generation have normal eyes and normal bodies. The students allow the flies in the F_1 generation to mate and produce an F_2 generation. The students then record observations for the flies in the F_2 generation and use the data to perform a chi-square goodness-of-fit test for a model of independent assortment. The setup for the chi-square goodness-of-fit test is presented in Table 1.

Table 1. Setup for the students' chi-square goodness-of-fit test

Phenotype	Observed	Expected
Normal eyes, normal body	231	279
Normal eyes, ebony body	86	93
Sepia eyes, normal body	97	93
Sepia eyes, ebony body	82	31

The students calculate a chi-squared value of 92.86 and compare it with a critical value of 7.82. Which of the following best completes the chi-square goodness-of-fit test?

(A) The null hypothesis cannot be rejected, and the students should conclude that the data fit a model of independent assortment.

(B) The null hypothesis cannot be rejected, and the students should conclude that the data may have resulted from genetic linkage.

(C) The null hypothesis can be rejected, and the students should conclude that the data fit a model of independent assortment.

(D) The null hypothesis can be rejected, and the students should conclude that the data may have resulted from genetic linkage. ✓

Answer D

Correct. Because the chi-square value is greater than the critical value, the null hypothesis can be rejected. The rejection of the null hypothesis indicates that the data do not fit a model of independent assortment. Consequently, the students can infer that the traits may be genetically linked.

80. Australian dragon lizards have a ZW sex-determination system. The male genotype is homogametic (ZZ), and the female genotype is heterogametic (ZW). However, all eggs incubated at temperatures above $32^\circ C$ tend to develop into females.

Which of the following best explains how the development of phenotypic female Australian dragon lizards with a ZZ genotype occurs when incubation temperatures are above $32^\circ C$?

5 Heredity MCQ

(A) Lizard embryos with a ZZ genotype cannot develop at temperatures above $32^{\circ}\mathrm{C}$.

(B) At incubation temperatures above $32^{\circ}\mathrm{C}$, Z chromosomes are mutated into W chromosomes.

(C) At incubation temperatures above $32^{\circ}\mathrm{C}$, crossing over transfers genes from the W chromosome to the Z chromosome, producing females.

(D) Incubation temperatures above $32^{\circ}\mathrm{C}$ inhibit the genes on the Z chromosome that produce proteins necessary for male development. ✓

Answer D

Correct. Similar in function to other SRY genes found on the Y chromosome in mammals, the Z chromosome contains genes that need to actively produce proteins during development of the embryo to trigger production of male characteristics. Above $32^{\circ}\mathrm{C}$, these genes are inhibited, so the default female phenotype develops.

81. A student carries out a genetics experiment with fruit flies to investigate the inheritance pattern of the white eye trait. The student crosses a homozygous white-eyed female with a wild-type male and records observations about the flies in the F_1 generation. The student plans to use the F_1 data to perform a chi-square goodness-of-fit test for a model based on an X-linked recessive pattern of inheritance. The student will use one degree of freedom and a significance level of $p = 0.05$. The setup for the student's chi-square goodness-of-fit test is presented in Table 1.

Table 1. Setup for the student's chi-square goodness-of-fit test

Phenotype	Observed	Expected
Red-eyed female	53	50
White-eyed male	47	50

The student calculates a chi-square value of 0.36. Which of the following statements best completes the student's chi-square goodness-of-fit test?

(A) The critical value is 0.05, and the student cannot reject the null hypothesis.

(B) The critical value is 0.05, and the student can reject the null hypothesis.

(C) The critical value is 3.84, and the student cannot reject the null hypothesis. ✓

(D) The critical value is 3.84, and the student can reject the null hypothesis.

Answer C

Correct. The critical value is 3.84. Because the calculated chi-square value is less than the critical value, the student cannot reject the null hypothesis. Consequently, the student may conclude that the data fit a

model in which the white eye trait has an X-linked recessive pattern of inheritance.

82. When a mustard plant seedling is transferred to an environment with higher levels of carbon dioxide, the new leaves have a lower stomata-to-surface-area ratio than do the seedling's original leaves.

Which of the following best explains how the leaves from the same plant can have different stomatal densities when exposed to an elevated carbon dioxide level?

(A) Increased photosynthesis leads to larger leaves that need more stomata for photosynthesis, leading to an increase in stomatal density.

(B) Leaf growth is promoted through increased photosynthesis, but the genetically regulated rate of stomatal production is not altered, leading to a decrease in stomatal density. ✓

(C) Leaf growth is inhibited by decreased photosynthesis, and the genetically regulated rate of stomatal production remains the same, leading to an increase in stomatal density.

(D) Leaf growth is inhibited by decreased photosynthesis, and the genetically regulated rate of stomatal production remains the same, leading to a decrease in stomatal density.

Answer B

Correct. Elevated carbon dioxide levels promote leaf growth due to increased photosynthetic activity. If the number of stomata produced remains the same as the leaves grow larger, the stomatal density decreases. This is an example of phenotypic plasticity, which is an adaptive trait.

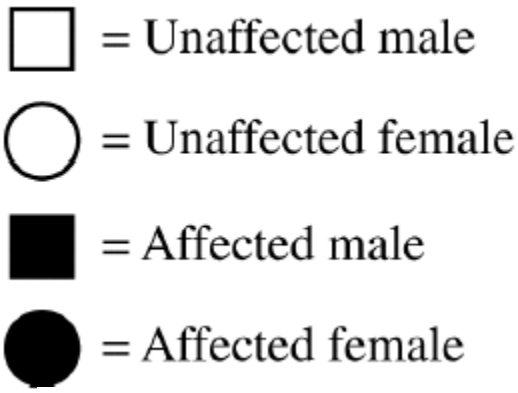
Tay-Sachs disease is a rare inherited disorder caused by an autosomal recessive allele of the *HEXA* gene. Affected individuals exhibit severe neurological symptoms and do not survive to reproductive age. Individuals who inherit one copy of the allele (Tay-Sachs carriers) typically show no symptoms of the disorder. The frequencies of Tay-Sachs carriers in the general population of North America and in three different subpopulations are presented in the table.

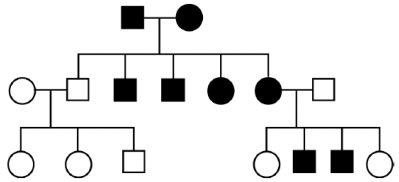
FREQUENCY OF TAY-SACHS CARRIERS IN DIFFERENT POPULATIONS

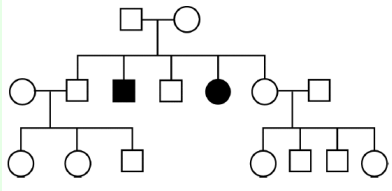
Population	Frequency of Tay-Sachs Carriers
General population	0.004
Subpopulation I	0.037
Subpopulation II	0.035
Subpopulation III	0.020

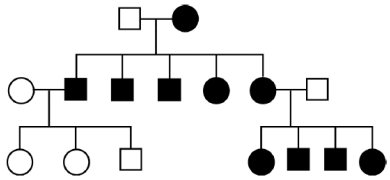
5 Heredity MCQ

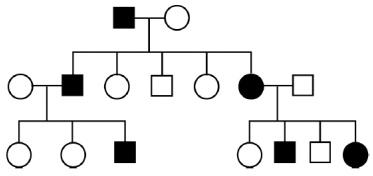
83. Which of the following pedigrees most accurately represents a family with a history of Tay-Sachs disease?



(A) 

(B)  ✓

(C) 

(D) 

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84. Himalayan rabbits are a breed of rabbits with highly variable fur color. If genetically similar rabbits are raised in environments that have different temperature conditions, the rabbits can have different color patterns.

Which of the following statements best explains how the fur color can be different in Himalayan rabbits raised under different temperature conditions?

5 Heredity MCQ

(A) The genotype does not contribute to coat color in Himalayan rabbits.

(B) The phenotype determines the genotype of coat color in Himalayan rabbits.

(C) Different environments cause specific mutations in the genes controlling pigment production.

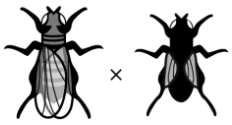
(D) The environment determines how the genotype is expressed. ✓

Answer D

Correct. The environment influences gene activity, which leads to a change in the phenotype of fur color. Temperature is the most significant environmental factor in determining the coat color of these rabbits.

85.

FRUIT FLY EXPERIMENTAL RESULTS



BbVv
Wild-Type Female
(tan body, wild-type wings)

×

bbvv
Mutant Male
(black body, vestigial wings)

Genotypes	*BbVv* Tan, Wild Type	*bbvv* Black, Vestigial	*Bbvv* Tan, Vestigial	*bbVv* Black, Wild Type
Expected results	862	862	862	862
Observed phenotypes	1,447	1,416	309	277

In *Drosophila melanogaster* the allele for wild-type tan body color (B) is dominant to the recessive allele for black body color (b). Similarly, the allele for wild-type wing shape (V) is dominant to the recessive allele for vestigial wing phenotype (v). In the cross diagrammed above, the expected and observed results are shown. Which of the following best explains the observed results of the cross?

(A) The alleles for body color and wing shape assort independently, as predicted by Mendel's laws.

(B) The genes for body color and wing shape are located close to each other on the same chromosome. ✓

(C) The traits of body color show complete dominance over the traits of wing shape.

(D) The observed variations in body color and wing shape are detectable in males but not in females

5 Heredity MCQ

86. A culture of white-eyed fruit flies (*Drosophila melanogaster*) was maintained for many generations. Females from the stock white-eyed culture were crossed with red-eyed (wild-type) males. The F_1 females were crossed with the white-eyed males from the original culture. The resulting phenotypes of the progeny are summarized below.

Parental Generation Cross	**F_1 Generation (at least 500 flies)**
White-eyed females × red-eyed males	100% of females are red-eyed 100% of males are white-eyed

F_1 Generation Cross	**F_2 Generation (at least 500 flies)**
F_1 red-eyed females × white-eyed males	50% of females are red-eyed and 50% are white-eyed 50% of males are red-eyed and 50% are white-eyed

There are white-eyed females in the F_2 generation because

(A) white is a dominant allele

(B) the white allele is autosomal

(C) a mutation has occurred

(D) these F_2 females have two white alleles ✓

(E) the white allele is located on the *Y* chromosome

87. The tiny blue-eyed Mary flower is often one of the first flowers seen in the spring in some regions of the United States. The flower is normally blue, but sometimes a white or pink flower variation is found.

The following data were obtained after several crosses.

Parents	F_1	F_2
Blue x white	Blue	196 blue, 63 white
Blue x pink	Blue	149 blue, 52 pink
Pink x white	Blue	226 blue, 98 white, 77 pink

Which of the following statements best explains the data?

5 Heredity MCQ

(A) The appearance of blue in the F_1 generation of the pink and white cross demonstrates that flower color is not an inherited trait but is determined by the environment.

(B) Flower color depends on stages of flower development, and young flowers are white, advancing to pink and then blue.

(C) Since the F_1 and F_2 phenotypes of the pink and white cross do not fit the expected genotypic and phenotypic ratios, blue-eyed Mary must reproduce by vegetative propagation.

(D) Flower color is an inherited trait, and the F_1 and F_2 phenotypes of the flowers arising from the pink and white cross can best be explained by another gene product that influences the phenotypic expression. ✓

Answer D

This option is correct. The student is asked to evaluate the data given. The crosses between blue / white and blue / pink show expected 3:1 ratios in the F_1 generation. The appearance of a majority of blue flowers in the F_1 cross between pink and white indicate that there is another gene product affecting the outcome.

88. The tiny blue-eyed Mary flower is often one of the first flowers seen in the spring in some regions of the United States. The flower is normally blue, but sometimes a white or pink flower variation is found.

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(D) Flower color is an inherited trait, and the F_1 and F_2 phenotypes of the flowers arising from the pink and white cross can best be explained by another gene product that influences the phenotypic expression. ✓

5 Heredity MCQ

89. The tadpoles of Mexican spadefoot toads are known to exhibit phenotypic plasticity depending on food availability. Tadpole mouthparts can vary significantly, prompting researchers to categorize them as either omnivore-morph or carnivore-morph. Carnivore-morph tadpoles are larger and have mouthparts that are better suited for predation. Remarkably, carnivore-morph tadpoles can change into omnivore-morph tadpoles when the food supply changes.

Which of the following best describes an advantage of the phenotypic plasticity displayed by the tadpoles?

(A) It allows the tadpoles to change their genome in response to environmental pressures.

(B) It enables the tadpoles to develop into a distinct species of toads.

(C) It gives the tadpoles increased versatility with respect to diet. ✓

(D) It allows the tadpoles to delay metamorphosis until there is maximal food available for the adults.

90. In a strain of tomato plants, short plant height and small fruit size are traits that display autosomal recessive patterns of inheritance. To investigate whether the traits segregate independently, researchers cross a pure-breeding line of tall tomato plants that have large fruits with a pure-breeding line of short tomato plants that have small fruits. The researchers observe that all the plants in the F_1 generation are tall and have large fruits. The researchers cross the F_1 plants with one another to generate an F_2 generation. The researchers record observations for the F_2 generation and will use the data to perform a chi-square goodness-of-fit test for a model of independent assortment. The setup for the chi-square goodness-of-fit test is shown in Table 1.

Table 1. Observed and expected counts of tomato plants in the F_2 generation

Phenotype	Observed	Expected
Tall plant with large fruits	104	90
Tall plant with small fruits	17	30
Short plant with large fruits	26	30
Short plant with small fruits	13	10

The researchers choose a significance level of $p = 0.05$. Which of the following best completes the chi-square goodness-of-fit test?

(A) The calculated chi-square value is 9.24, and the critical value is 7.82. The null hypothesis of independent assortment can be rejected. ✓

(B) The calculated chi-square value is 9.24, and the critical value is 9.49. The null hypothesis of independent assortment cannot be rejected.

(C) The calculated chi-square value is 13.13, and the critical value is 7.82. The null hypothesis of independent assortment can be rejected.

(D) The calculated chi-square value is 13.13, and the critical value is 9.49. The null hypothesis of independent assortment cannot be rejected.

5 Heredity MCQ

Answer A

Correct. The calculated chi-square value is 9.24. The critical value for a chi-square test with 3 degrees of freedom and a significance level of $p = 0.05$ is 7.82. Because the calculated chi-square value is greater than the critical value, the null hypothesis of independent assortment can be rejected. The researchers can conclude that the data do not fit a model of independent assortment and that the traits may be genetically linked.

91. A culture of white-eyed fruit flies (*Drosophila melanogaster*) was maintained for many generations. Females from the stock white-eyed culture were crossed with red-eyed (wild-type) males. The F_1 females were crossed with the white-eyed males from the original culture. The resulting phenotypes of the progeny are summarized below.

Parental Generation Cross	**F_1 Generation (at least 500 flies)**
White-eyed females × red-eyed males	100% of females are red-eyed 100% of males are white-eyed

F_1 Generation Cross	**F_2 Generation (at least 500 flies)**
F1 red-eyed females × white-eyed males	50% of females are red-eyed and 50% are white-eyed 50% of males are red-eyed and 50% are white-eyed

Which of the following best describes the mode of inheritance of eye color in the white culture?

(A) Autosomal

(B) Dominant

(C) Located on the *Y* chromosome

(D) Sex-linked ✓

(E) Lethal

5 Heredity MCQ

92. Students carry out a genetics experiment to investigate the inheritance pattern of the white-eye trait in fruit flies. In the experiment, the students cross a red-eyed female with a white-eyed male to produce an F_1 generation. The students observe that all the flies in the F_1 generation have red eyes. The students then allow the F_1 flies to mate and produce an F_2 generation. The students will use the F_2 data to perform a chi-square goodness-of-fit test based on a null hypothesis of autosomal recessive inheritance. The observed and expected values for the chi-square goodness-of-fit test are shown in Table 1.

Table 1. The observed and expected values for a chi-square goodness-of-fit test

Phenotype	Observed	Expected
Red-eyed female	497	375
Red-eyed male	254	375
White-eyed female	0	125
White-eyed male	249	125

The students plan to use a significance level of $p = 0.01$. Which of the following is the most appropriate critical value for the students to use in their chi-square goodness-of-fit test?

(A) 7.82

(B) 11.34 ✓

(C) 13.28

(D) 326.7

Answer B

Correct. The critical value for a significance level of $p = 0.01$ and 3 degrees of freedom is 11.34. A calculated chi-square value greater than 11.34 will allow the students to reject the null hypothesis of autosomal recessive inheritance.

93. A human cell in prophase of mitosis can be distinguished from a human cell in prophase I of meiosis by the presence of

(A) kinetochores in the mitotic cell

(B) a spindle in the mitotic cell

(C) twice as many chromosomes in the meiotic cell

(D) half as many chromosomes in the meiotic cell

(E) paired homologous chromosomes in the meiotic cell ✓

5 Heredity MCQ

94. For sexually reproducing diploid parent cells, which of the following statements best explains the production of haploid cells that occurs in meiosis but not in mitosis?

(A) Separation of chromatids occurs once, and there is one round of cell division in meiosis.

(B) Separation of chromatids occurs twice, and there are two rounds of cell division in mitosis.

(C) Separation of chromatids occurs once, and there are two rounds of cell division in meiosis. ✓

(D) Separation of chromatids occurs twice, and there is one round of cell division in mitosis.

Answer C

Correct. Separation of chromatids occurs once in meiosis, and there are two rounds of cell division, which ensures that haploid gamete cells are formed in sexually reproducing diploid organisms.

95. Humans have a diploid number ($2n$) of 46. Which of the following statements best predicts the consequence if meiosis did not occur during gametogenesis?

(A) The gametes would get larger from one generation to the next.

(B) The chromosome number would double with each generation. ✓

(C) The chromosome number would be halved with each generation.

(D) The chromosome number would triple with each generation.

96. Which of the following best describes the cells that result from the process of meiosis in mammals?

(A) They are diploid.

(B) They can be used to repair injuries.

(C) They are genetically different from the parent cell. ✓

(D) They are genetically identical to all the other cells in the body.

(E) They are identical to each other.

97. All of the following are true statements about meiosis in mammals EXCEPT:

(A) It serves as a factor in bringing about variation among offspring.

(B) It follows DNA replication.

(C) It occurs only in reproductive structures.

(D) It produces cells with the haploid number of chromosomes.

(E) It produces four genetically identical gametes. ✓

5 Heredity MCQ

98. Table 1 shows the stage and number of cells and chromosomes per cell at the end of the stage in a $2n = 24$ organism.

Table 1. Cell and chromosome count during selected phases of meiosis

Stage	Number of Cells	Number of Chromosomes per Cell
Prophase I	1	24
Metaphase I	1	24
Anaphase I	1	24
Telophase I	1	24
Beginning of Prophase II	2	12

Which of the following statements correctly describes the chromosomes in each daughter cell at the end of meiosis I ?

(A) Each daughter cell contains 12 chromatids. Each chromatid is one of two from a single chromosome with the other one of the pair found in the other daughter cell.

(B) Each daughter cell contains 12 chromosomes, each composed of two chromatids. Since the chromosomes were randomly divided, one daughter cell may contain both of a pair of homologous chromosomes, while the other cell contains both of another pair of homologous chromosomes.

(C) Each daughter cell contains 12 chromosomes, each composed of two chromatids. Each chromosome is one of a pair of homologous chromosomes from the parent cell, with the other homologue found in the other daughter cell. ✓

(D) Each daughter cell contains 24 separate chromatids. Since every two chromatids were originally joined, forming one homologous chromosome, the number of chromatids is divided by two to determine the number of chromosomes.

Answer C

Correct. The described distribution of chromosomes ensures that each daughter cell contains a complete haploid genome. Each chromosome still consists of two chromatids, formed during the S phase of the cell cycle. Chromatids will separate into two daughter cells during meiosis II.

99. Both mitosis and meiosis begin with a parent cell that is diploid. Which of the following best describes how mitosis and meiosis result in daughter cells with different numbers of chromosomes?

5 Heredity MCQ

(A) In mitosis, the chromosomes consist of a single chromatid, which is passed to two haploid daughter cells. In meiosis, the chromosomes consist of two chromatids during the first round of division and one chromatid during the second round of division, resulting in two haploid daughter cells.

(B) In mitosis, synapsis of homologous chromosomes results in four haploid daughter cells after one division. In meiosis, synapsis of homologous chromosomes occurs during the second division and results in four diploid daughter cells.

(C) Mitosis produces one identical daughter cell after one round of division. Meiosis has two rounds of division and doubles the number of chromosomes in the second round of division, producing four diploid cells.

(D) Mitosis produces two identical diploid daughter cells after one round of division. Meiosis produces four haploid daughter cells after two rounds of division. ✓

Answer D

Correct. The function of mitosis is to produce identical daughter cells, while the function of meiosis is to produce gametes. Meiosis I reduces the chromosome number, and meiosis II separates the sister chromatids in the remaining chromosomes.

100. Which of the following best explains a distinction between metaphase I and metaphase II?

(A) The nuclear membrane breaks down during metaphase I but not during metaphase II.

(B) Chromosomes align at the equator of the cell during metaphase II but not during metaphase I.

(C) The meiotic spindle is needed during metaphase I but not during metaphase II.

(D) Homologous pairs of chromosomes are aligned during metaphase I, but individual chromosomes are aligned during metaphase II. ✓

Answer D

Correct. During metaphase I, tetrads (homologous pairs of chromosomes) are aligned along the metaphase plate. During metaphase II, however, single chromosomes (each composed of two chromatids) are aligned along the metaphase plate.

101. Which of the following occurs in both meiosis and mitosis?

(A) Separation of homologous chromosomes

(B) Reduction of chromosome number

(C) Separation of sister chromatids ✓

(D) Crossing-over between homologous chromosomes

(E) Production of four daughter cells

5 Heredity MCQ

102. Which of the following occurs during mitosis but not during meiosis I ?

(A) The chromosomes are pulled to opposite poles of the spindle apparatus.

(B) The chromatids of each chromosome are separated. ✓

(C) The nuclear envelope breaks down.

(D) Both synapsis and crossing-over take place.

(E) The diploid number of chromosomes is reduced to the haploid number.

103. *Saccharomyces cerevisiae* is a diploid yeast species that can reproduce either sexually or asexually. An experiment was performed to induce mitotically dividing *S. cerevisiae* cells in G_2 to undergo meiosis. Which of the following best describes the steps these cells will follow to form gametes?

(A) The first division will result in crossing over between homologous chromosomes, and the second division will reduce the original number of chromosomes by half in the daughter cells.

(B) The first division will reduce the number of chromosomes by half for each daughter cell, and the second division will result in each daughter cell having one-fourth of the original number of chromosomes.

(C) The first division will move single chromatids to each daughter cell, and the second division will double the number of chromosomes in each daughter cell.

(D) The first division will reduce the number of chromosomes by half for each daughter cell, and the second division will move single chromatids to each daughter cell. ✓

Answer D

Correct. The homologous pairs of chromosomes separate in meiosis I, resulting in two haploid daughter cells. The chromosomes had replicated prior to G_2 and consist of two sister chromatids. The chromatids separate during meiosis II, resulting in a total of four haploid cells.

5 Heredity MCQ

104.

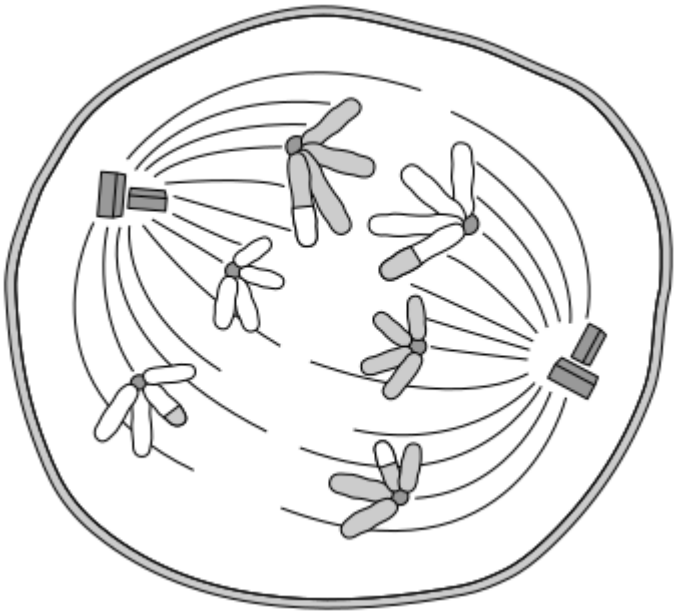


Figure 1. A cellular process

Which of the following best describes the cellular process illustrated in Figure 1 ?

(A) Sister chromatids separating during anaphase of mitosis

(B) Chromosomes lining up along the midline of the cell during mitosis

(C) Reducing the chromosome number during anaphase I of meiosis ✓

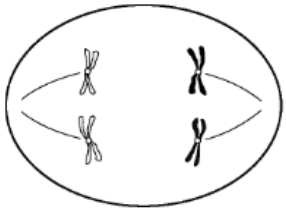
(D) Chromatids failing to separate during meiosis

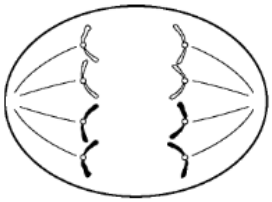
Answer C

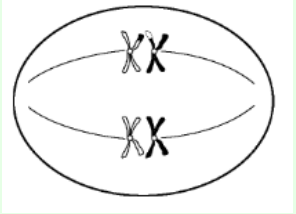
Correct. The fact that each chromosome is composed of a pair of chromatids, some of the chromatids have undergone crossing-over, indicates that these are homologous chromosomes in meiosis. The movement of the chromosomes toward opposite poles supports the claim that this is an illustration of anaphase I.

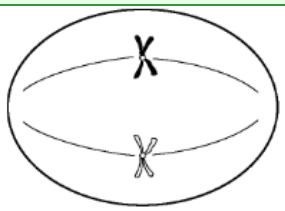
105. Within a forest ecosystem, there is a large amount of diversity among members of a warbler species. Of the following stages of meiosis illustrated for a typical cell, which contributes most to diversity among the warblers?

5 Heredity MCQ

(A) 

(B) 

(C)  ✓

(D) 

106. In most vertebrates, the sperm cell normally contributes which of the following to the new organism?

(A) Many mitochondria

(B) Significant amounts of RNA

(C) A haploid complement of chromosomes ✓

(D) Most of the cytoplasm of the zygote

(E) Two sex chromosomes

5 Heredity MCQ

107.

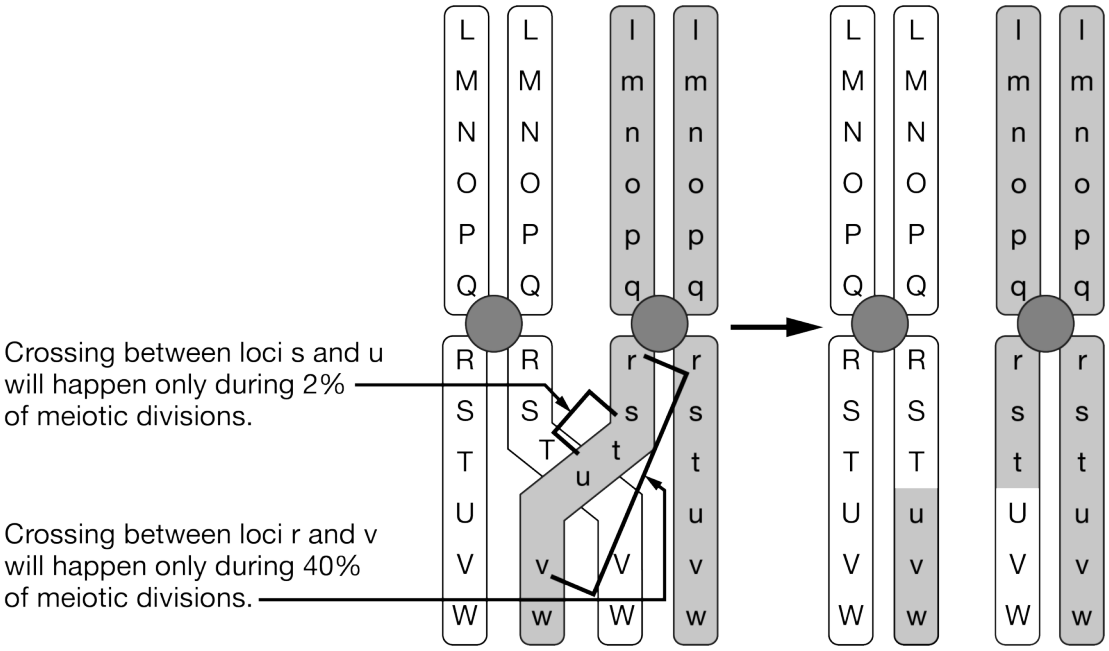


Figure 1. Model of crossing over between homologous chromosomes, indicating crossing over rate of selected loci.

During prophase I replicated homologous chromosomes pair up and undergo synapsis. What testable question is generated regarding synapsis and genetic variability by Figure 1 ?

(A) Is the distance between two gene loci related to crossover rate? ✓

(B) Does crossing over occur more often in some chromosomes than in others?

(C) Is crossing over inhibited by methylation?

(D) Is crossing over promoted by methylation?

Answer A

Correct. During meiosis I, homologous, non-sister chromatids exchange genetic material via a process called crossing over (recombination). The likelihood that a recombination event will separate two genes on the same chromosome increases as the distance between the two gene loci increases.

5 Heredity MCQ

108. A model of crossing over during gamete formation is shown in Figure 1.

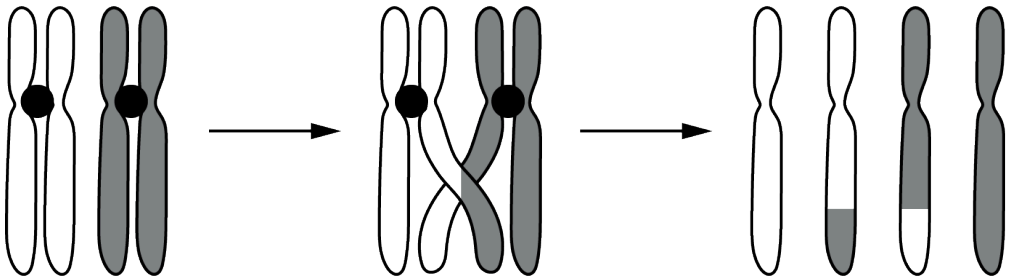


Figure 1. Model of crossing over during meiosis

Based on Figure 1, which of the following questions could best be addressed?

(A) Does synapsis of homologous chromosomes in the parent cell contribute to an increase in genetic diversity in the daughter cells? ✓

(B) Do sister chromatids separate and form diploid daughter cells?

(C) Do chromatids from nonhomologous chromosomes rearrange to produce identical daughter cells?

(D) Does synapsis of nonhomologous chromosomes produce daughter cells that are identical to the parent cell?

Answer A

Correct. Based on the model, synapsis of homologous chromosomes allows the opportunity for crossing over to occur, which can result in chromosomal rearrangement and increased genetic variation in the daughter cells.

109. Scientists have found that DNA methylation suppresses crossing-over in the fungus *Ascobolus immersus*. Which of the following questions is most appropriately raised by this specific observation?

(A) Is the level of genetic variation in the gametes related to the amount of DNA methylation observed? ✓

(B) Without crossing-over, will gametes be viable and be able to produce zygotes?

(C) Does DNA methylation result in shorter chromosomes?

(D) Is this species of fungus a diploid organism?

Answer A

Correct. DNA methylation is shown to suppress crossing-over, which is an important source of genetic

5 Heredity MCQ

variation. Thus, it is reasonable to investigate how the level of genetic variation in gametes being produced is related to how much DNA has been methylated.

110. In anaphase I of meiosis, cohesion between the centromeres of sister chromatids is maintained while homologous chromosomes migrate to opposite poles of the cell along the meiotic spindle as represented in Figure 1.

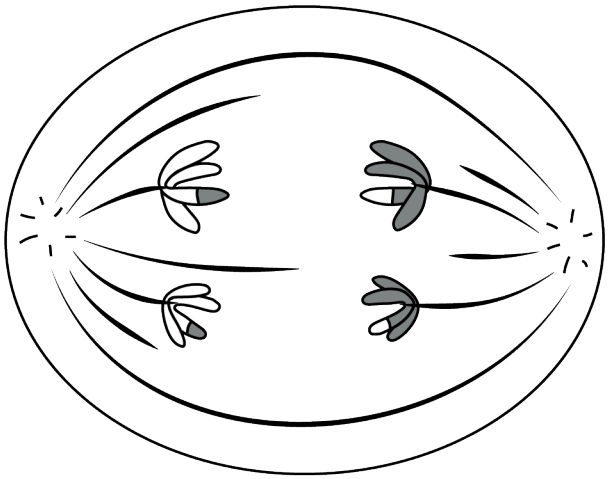


Figure 1. Migration of homologous chromosomes during anaphase I of meiosis

A compound that prevents the separation of the homologous chromosomes in anaphase I is being studied. Which of the following questions can be best answered during this study?

(A) Will the cells produced at the end of meiosis still be genetically identical to each other in the presence of this compound?

(B) Will the long-term development of the individual be affected by this meiotic error?

(C) When do the centrosomes start to move apart during meiosis I as compared to meiosis II?

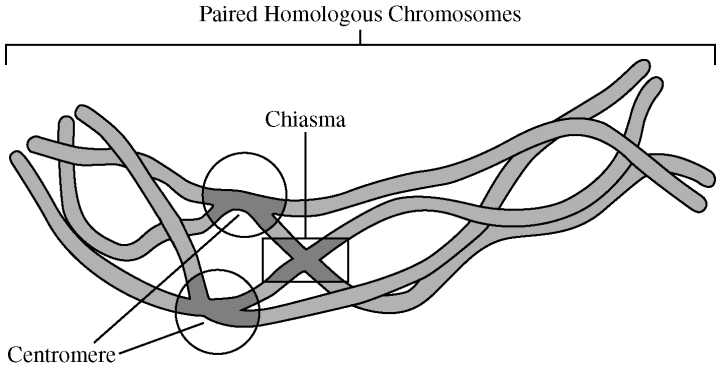
(D) Is there a pattern to the movement of homologous chromosomes in the presence of this compound? ✓

Answer D

Correct. The compound prevents separation of homologous chromosomes, meaning that both the maternal and paternal chromosomes will migrate along the spindle fibers together. Patterns and other aspects of the movement of these homologous chromosomes in the presence of this compound could be recognized effectively during this study.

5 Heredity MCQ

111.



The process depicted in the image above is best summarized by which of the following descriptions?

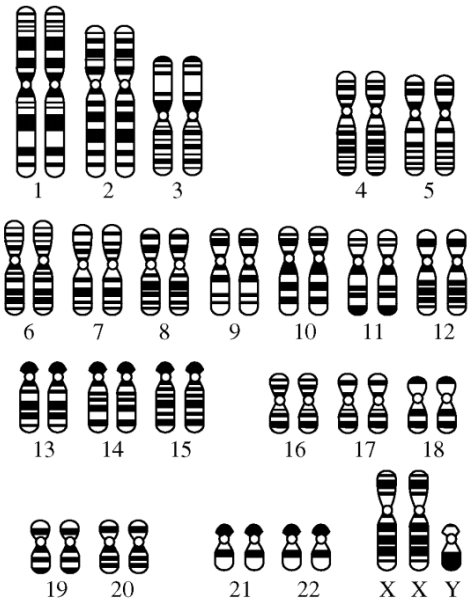
(A) During the synthesis phase of the cell cycle, DNA molecules replicate to generate identical daughter cells.

(B) Centromeres align specific gene sequences of homologous chromosomes during mitotic divisions.

(C) The spindle apparatus attaches at chiasma during metaphase of mitosis.

(D) During meiosis, crossing over leads to recombination of alleles between homologous chromosomes. ✓

112.



The diagram above depicts a karyotype of an individual human. Which of the following statements concerning the karyotype in the diagram is true?

(A) The diagram illustrates a genetic condition found in females.

(B) The diagram indicates a mechanism for increasing genetic diversity in subsequent generations.

(C) The diagram illustrates the results of nondisjunction during gamete formation. ✓

(D) The diagram indicates Down syndrome, a genetic condition.

5 Heredity MCQ

113. A model showing two possible arrangements of chromosomes during meiosis is shown in Figure 1.

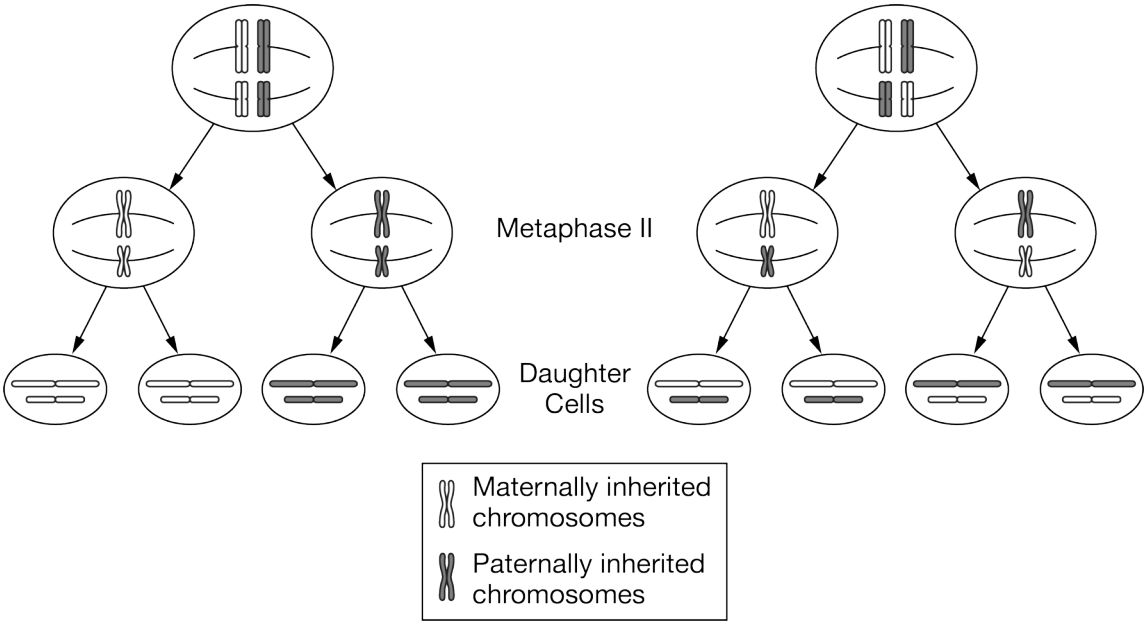


Figure 1. Two possible arrangements of chromosomes during meiosis

Which of the following questions about genetic diversity could most appropriately be answered by analysis of the model in Figure 1 ?

(A) Does crossing-over generate more genetic diversity than the fusion of gametes does?

(B) Does DNA methylation prevent independent assortment during metaphase II?

(C) How does the independent assortment of the two sets of homologous chromosomes increase genetic diversity? ✓

(D) Do daughter cells that are not genetically identical to parent cells produce viable zygotes?

Answer C

Correct. By mixing the maternally and paternally inherited chromosomes in the parent and passing different combinations on to each daughter cell or gamete, the number of combinations of alleles in the gametes is increased. Thus, the gametes have a greater potential to increase genetic diversity in future generations.

114. If 2n = 48 for a particular cell, then the chromosome number in each cell after meiosis would be

5 Heredity MCQ

(A) 96
(B) 48
(C) 24 ✓
(D) 12
(E) 6

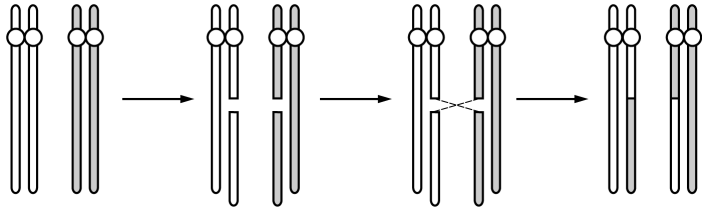
115. The events listed below generally take place during meiosis.

I. Synapsis occurs.
II. Crossing-over is completed.
III. Condensation of chromosomes begins.
IV. Separation of homologous chromosomes begins.

Which of the following is the correct sequence of these events?

(A) I, II, III, IV
(B) II, I, III, IV
(C) III, I, II, IV ✓
(D) III, IV, II, I
(E) IV, II, III, I

116.



The diagram above illustrates which of the following processes?

(A) Crossing-over ✓
(B) Base pair substitution
(C) Duplication
(D) Deletion
(E) Posttranscriptional processing

117. In a breeding pair of eukaryotic organisms, no crossing over occurs between chromatids of homologous chromosome pairs during either egg formation or sperm formation. Which of the following best explains the genetic makeup of the offspring produced when eggs are fertilized by sperm in this breeding pair?

5 Heredity MCQ

(A) All offspring will be genetically identical to the parent that has the greatest number of dominant traits.

(B) Half of the offspring will be genetically identical to one parent and the other half will be genetically identical to the other parent.

(C) All offspring will be genetically different from each other and from the parents. ✓

(D) The zygotes from fertilization will not develop into any offspring because of nondisjunction.